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Inhibition of mucin-type *O*-glycosylation impairs melanogenesis, melanoma growth, and metastatic capacity

Tanya Jain[1#], Abhiraj Rajasekharan[1#], Simran Gorai[1], Nagarajan Perumal[2], Santiswarup Singha[3], Srinivasa-Gopalan Sampathkumar[1*]

[1]Laboratory of Chemical Glycobiology, [2]Experimental Animal Facility, [3]Nanobiotechnology Laboratory, National Institute of Immunology (NII), Aruna Asaf Ali Marg, New Delhi 110067, India

*Corresponding author; email: gopalan@nii.ac.in

[#]These authors contributed equally

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Summary

Mucin-type *O*-glycans on melanoma cell surfaces and the melanosomal protein, Pmel17/gp100, regulate melanogenesis and melanoma metastasis. The *O*-glycosylation inhibitor, peracetyl *N*-thioglycolyl-D-galactosamine (Ac_5GalNTGc, **1a**), suppressed melanin synthesis and induced hyposialylation, thereby disrupting Siglec-mediated immune evasion and retarding tumor growth and metastasis.

Abstract

Tumor-associated glycans are critical regulators of immune evasion in melanoma. Despite the success of immune checkpoint blockade, resistance and relapse remain a challenge, highlighting the need to therapeutically target additional immunosuppressive axes. One such axis involves Siglec-sialoglycan interactions that facilitate tumor immune evasion. The melanoma-associated antigen Pmel17/gp100 is a melanosomal glycoprotein overexpressed in tumor cells and essential for melanosomal architecture. Pmel17/gp100 forms amyloid fibrils within melanosomes, and its extensive *O*-glycosylation contributes to melanin biosynthesis and melanoma progression. We show that pharmacological inhibition of *O*-glycosylation using peracetyl *N*-thioglycolyl-D-galactosamine (Ac_5GalNTGc, **1a**) decreases Pmel17/gp100 glycosylation and melanin synthesis in B16F10 melanoma cells, induces tumor surface hyposialylation, diminishes Siglec-E engagement, and disrupts the Siglec-sialoglycan immune checkpoint. In C57BL/6J mice, **1a** significantly delayed tumor growth, prolonged survival, and reduced lung metastases relative to controls. These findings identify *O*-glycosylation as a dual-mode therapeutic vulnerability, linking melanosome dysfunction to attenuation of Siglec-mediated immune evasion in melanoma.

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Introduction

Specific molecular processes, regulated both spatially and temporally in a highly orchestrated manner, define the development and homeostasis of multicellular organisms from the zygote to adulthood. Rarely do these processes go awry at any stage, from birth to adulthood. However, the probability of disrupted homeostasis, dysregulated differentiation, an exhausted immune system, epithelial-to-mesenchymal transitions (EMT), and malignant transformation increases exponentially with age (De Magalhães, 2025). These changes necessitate pharmacological and clinical interventions, including curative therapies, immunotherapies, and vaccines. Among these pathologies, cancer represents an example of how dysregulated developmental and immune processes drive lethal disease.

Skin cancers, particularly melanoma, represent one of the most aggressive forms of cancer, characterized by high metastatic potential and poor prognosis at advanced stages. Despite recent advances in immunotherapy targeting PD-1/PD-L1 and CTLA-4/B7 immune checkpoints, as well as kinase inhibitors targeting BRAF V600E mutations, melanoma remains a significant clinical challenge with substantial morbidity and mortality (Schadendorf *et al.*, 2018). This is attributed to resistance, relapse, and metastases of melanoma. Increasing evidence suggests that melanogenesis and melanin accumulation contribute to melanoma aggressiveness. In most cases, increased melanin content positively correlates with disease severity and enhanced metastatic potential, emphasizing melanogenesis as a critical yet underexplored axis in melanoma biology (Cabaço *et al.*, 2022).

Melanogenesis occurs within melanosomes, specialized intracellular organelles in melanocytes characterized by a unique, highly organized amyloid fibrillar architecture, which is essential for melanin synthesis and transfer to keratinocytes (Wasmeier *et al.*, 2008). Melanosomes contain Pmel17/gp100 (pre-melanosomal protein) and MART-1/Melan-A within a limiting bilayer membrane and luminal compartment, where polymers of melanin (eumelanin and pheomelanin) are synthesized through tyrosinase-driven oxidation of tyrosine and DOPA. These enzymatic reactions involve highly reactive quinone intermediates and free radical species, necessitating tight spatial confinement to prevent cytotoxic damage. The melanosomal membrane, therefore, serves as a protective compartment, housing these radical-generating reactions and safeguarding the cytosol from oxidative stress. The structural protein Pmel17/gp100 plays a central role in melanosome biogenesis by forming organized amyloid-like fibrils that serve as scaffolds for melanin polymerization, facilitating efficient radical coupling and deposition while minimizing uncontrolled diffusion of reactive intermediates (Raposo and Marks, 2007).

It is known that Pmel17/gp100 is synthesized as a precursor polypeptide chain, extensively glycosylated, and subjected to multiple proteolytic cleavages prior to fibril formation through spontaneous molecular self-assembly. Both the polycystic kidney disease (PKD) and repeat (RPT) domains of Pmel17/gp100 are critical for maintaining the fibrillar architecture of melanosomes (Graham *et al.*, 2019). Notably, proteins containing mucin-like domains are intrinsically predisposed to spontaneous higher-order self-assembly and polymer formation due to their repetitive sequences and dense *O*-glycosylation, which influence structural organization and intermolecular interactions (Garbarino *et al.*, 2024). This structural propensity likely contributes to the

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controlled amyloidogenesis of Pmel17/gp100 within melanosomes. Importantly, melanoma frequently overexpresses melanocyte-specific proteins known as melanoma-associated antigens, including PMEL/gp100, tyrosinase, tyrosinase-related proteins 1 and 2 (TYRP1 and 2) (Naveh, Rao, and Butterfield, 2013). Notably, the majority of melanosomal proteins, including Pmel17/gp100, tyrosinase, TYRP1, and TYRP2, are glycoproteins; however, only a few studies have explored glycosylation as a therapeutic target for melanoma, hyperpigmentation, or depigmentation disorders, such as vitiligo.

Aberrant glycosylation is a hallmark of cancer, contributing to tumor growth, survival, invasion, and immune evasion (Pinho and Reis, 2015). Tumor cells frequently exhibited altered mucin-type *O*-glycosylation (MTOG) patterns, often characterized by increased expression of truncated *O*-glycans such as the sialyl-Tn-antigen (GalNAcα-Ser/Thr), as well as altered core structures and increased terminal sialylation (Kudelka *et al.*, 2015). Many of these aberrantly expressed glycans are recognized by monoclonal antibodies as tumor-associated carbohydrate antigens (TACAs) (Hakomori, 1989). Such glycosylation changes promote tumor progression through multiple mechanisms, including enhanced cell adhesion, resistance to apoptosis, modulation of anti-tumor immune responses, and masking of immunogenic tumor epitopes from cytotoxic T lymphocyte (CTL) recognition through altered antigen processing and peptide-MHC presentation (Harding *et al.*, 1993). Notably, many of these tumor-associated glycans are sialylated structures that can engage immune-inhibitory receptors.

Recently, the Siglec-sialoglycan axis has emerged as a critical immune checkpoint in cancer (Läubli and Varki, 2020). Siglecs (sialic acid-binding immunoglobulin-like lectins), expressed on immune cells, bind to sialylated glycans on the tumor cells. Upon engagement, Siglecs transduce immunosuppressive signals through immunoreceptor tyrosine-based inhibitory motifs (ITIM), thereby dampening anti-tumor immune responses. Beyond conventional immune checkpoint therapies using anti-PD-1 and anti-CTLA-4 antibodies, efforts are currently underway for the targeted removal of sialic acids from tumor cell sialoglycans using antibody-enzyme conjugates (AEC) (Gray *et al.*, 2020). Büll and coworkers have reported that glycomimetics or sialomimetics, particularly halogen-substituted derivatives of *N*-acetyl-D-neuraminic acid (NeuAc), show efficacy in retarding melanoma tumor growth in mice (Büll *et al.*, 2013). Esko and coworkers reported inhibitors of *O*-glycosylation using decoy mechanisms (Brown *et al.*, 2003). However, both approaches target only the elaboration of mucin-type *O*-glycan assembly, leaving the initiation steps of mucin-type *O*-glycosylation largely unexplored.

Analogues of *N*-acetyl-D-hexosamine (HexNAc) have been widely used for the manipulation of glycoconjugates in living systems through the HexNAc salvage pathways (Pouilly *et al.*, 2012). On the one hand, HexNAc analogues carrying keto-, azido-, and alkyne-functionalities have been employed for metabolic glycan engineering (MGE) (Almaraz *et al.*, 2012). On the other hand, HexNAc analogues carrying halogen groups have been shown to inhibit glycosylation without metabolic incorporation (Marathe *et al.*, 2010).

Our laboratory previously demonstrated that peracetyl *N*-thioglycolyl-D-galactosamine (Ac_5GalNTGc, **1a**) is an efficient inhibitor of MTOG, both *in vitro* and *in*

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vivo (Agarwal *et al.*, 2013; Dwivedi *et al.*, 2018; Wang *et al.*, 2021). **1a** is metabolically incorporated into glycoproteins and inhibits both glycan site occupancy of the Tn antigen, the first step in MTOG biosynthesis, and subsequent elaboration. Based on these findings, we hypothesized that small molecule inhibitors of MTOG that induce tumor hyposialylation would disrupt the Siglec-sialoglycan axis and enhance anti-tumor immunity (**Fig. 1A**). In melanoma, such an inhibitor could exert a dual therapeutic effect by (**a**) impairing melanosome structural integrity through defective *O*-glycosylation of Pmel17/gp100, thereby impairing melanin synthesis, and (**b**) decreasing sialoglycan expression on the tumor cell surface, thus relieving Siglec-mediated immune suppression.

Herein, we report studies demonstrating that pharmacological inhibition of *O*-glycosylation with **1a** profoundly alters melanoma cell biology both *in vitro* and *in vivo.* We show that treatment with **1a** decreased the Pmel17/gp100 glycosylation, decreased melanin content, and induced global hyposialylation, leading to impaired tumor cell migration and invasion in B16F10 (mouse melanoma) and B16F10-Luc2 (luciferase-expressing mouse melanoma) cells. In a syngeneic mouse melanoma model, **1a** significantly retarded tumor growth, reduced lung metastasis, and prolonged survival. Collectively, these findings establish *O*-glycosylation as a promising therapeutic target in melanoma and support the development of carbohydrate-based small-molecule inhibitors as combinatorial therapy in both adjuvant and neoadjuvant clinical settings.

Results

Mucin-type *O*-glycans are found abundantly in mucins and mucin-domain glycoproteins (MDGP) (e.g., CD43, CD162/PSGL-1); MTOG is also found in small clusters in the linker regions on several membrane and secreted glycoproteins (e.g., IgM, IgA) (Goth *et al.*, 2023). Biosynthesis of MTOG is initiated in the Golgi apparatus by the action of a family of polypeptide GalNAc transferases (ppGalNAcT), utilizing the UDP-GalNAc as an activated sugar-nucleotide donor, and resulting in the formation of α-GalNAc-Ser/Thr (the Tn-antigen) on proteins of the secretory pathway (Varki *et al.*, 2022) (**Supporting Figure 1A**). The Tn-antigen is elaborated to Thomson-Friedenreich antigen (TF- or T-antigen), sialyl-Tn, and disialyl-T structures by multiple galactosyl- and sialyl-transferases, in a competitive manner. Several reports have shown aberrant *O*-glycosylation in cancer, including melanoma, and the glycosylation pattern varies with disease stage, spanning primary tumors, circulating metastatic cells, and established metastatic cancer in a secondary tissue (Friedman *et al.*, 2012).

Mucins and MDGPs provide a dense sialic acid-rich glycocalyx on tumor cells, which helps in dampening the native tumor surveillance and elimination mechanism through interaction with Siglecs expressed on tumor-infiltrating effector T-cells (van de Wall *et al.*, 2020). The Siglec-sialoglycan axis renders the tumor microenvironment immunosuppressive, enabling tumor growth. Efforts to disrupt this immune checkpoint using antibody-neuraminidase conjugates and glycomimetic inhibitors have been the focus of recent research (Xiao *et al.*, 2016). Targeting glycan biosynthetic pathways upstream using small-molecule inhibitors represents a complementary strategy to interrupt this axis.

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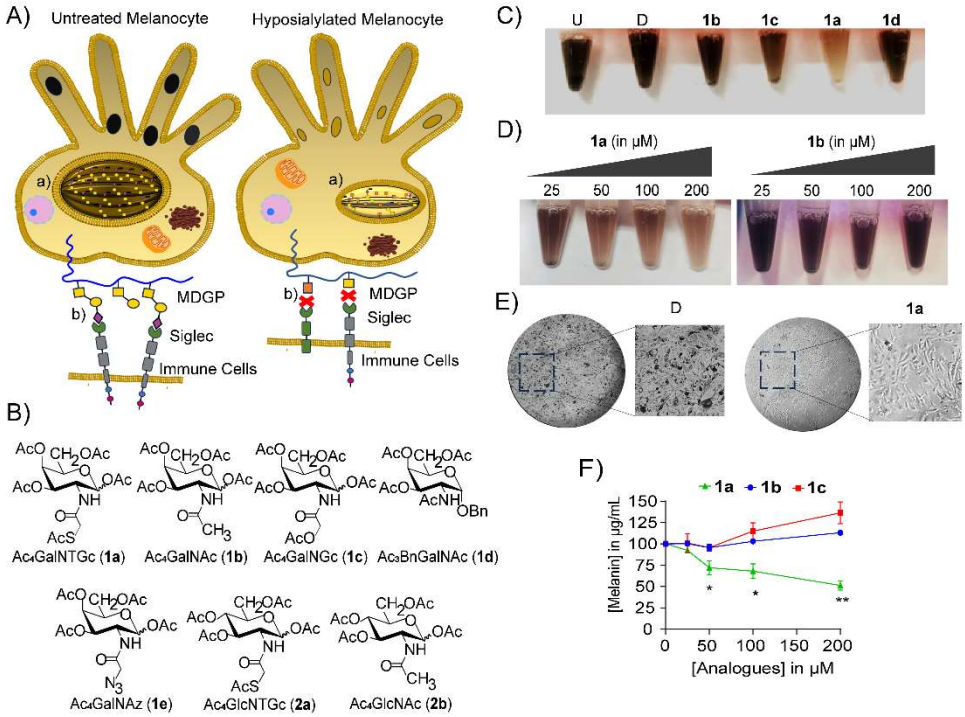


Figure 1. Metabolic glycan engineering using modified GalNAc analogues disrupts melanin biosynthesis. (A) Schematic model illustrating the role of *O*-glycosylation in melanosome structure and melanogenesis. Under physiological conditions, melanosomes contain organized amyloid-like fibrils of Pmel17/gp100 that provide a scaffold for melanin deposition and confer ellipsoid, sheet-like architecture. (a) Inhibition of *O*-glycosylation by **1a** disrupts melanosome integrity and melanogenesis, leading to defective melanosomes and decreased pigmentation. In cancer cells, hypersialylation promotes immune evasion via Siglec engagement; **1a**-induced hyposialylation may attenuate this Siglec-sialic acid axis. (B) Chemical structures of GalNAc and GlcNAc analogues used in this study: Ac_5GalNTGc (**1a**), featuring a thioglycolyl group in the *N*-acyl moiety; Ac_4GalNAc (**1b**), the peracetylated form of the wild-type GalNAc analogue; Ac_5GalNGc (**1c**), featuring a glycolyl group in the *N*-acyl moiety; Ac_3BnGalNAc (**1d**), an *O*-glycan inhibitor distinguished by its benzyl group; Ac_4GalNAz (**1e**), featuring an azide moiety; Ac_4GlcNAc (**2b**), the peracetylated form of the wild-type GlcNAc analogue; and Ac_5GlcNTGc (**2a**), the C4 epimer of **1a**. (C) Visual pigmentation of B16F10 cells suspended in 1× PBS after treatment with GalNAc analogues (100 μM, 48 h). (D) Dose-dependent pigmentation analysis of B16F10 cells suspended in 1× PBS treated with **1a** or **1b** (25-200 μM, 48 h). (E) Phase-contrast images (10×) of B16F10 cells treated with D (vehicle) or **1a**, showing intracellular melanin granules. (F) Quantification of melanin content following treatment with increasing concentrations (25, 50, 100, and 200 μM) of GalNAc analogues. Data represent mean ± SD (standard deviation) of three biological replicates. Statistical significance was determined by using one-way analysis of variance (ANOVA). *P <0.05, **P <0.01 vs. D). U, untreated; and D, DMSO (vehicle). Monosaccharide symbols follow the Symbol Nomenclature for Glycans (SNFG)(Neelamegham *et al.*, 2019).

To investigate the structure-activity relationship, we employed a panel of GalNAc analogues, *viz.*, peracetyl *N*-thioglycolyl-D-galactosamine (Ac_5GalNTGc, **1a**, thiol-containing analogue) (Du *et al.*, 2011), peracetyl *N*-acetyl-D-galactosamine (Ac_4GalNAc, **1b**, wild-type control) (Agarwal *et al.*, 2013), peracetyl *N*-glycolyl-D-galactosamine (Ac_5GalNGc, **1c**, *N*-glycolyl analogue) (Agarwal *et al.*, 2013), peracetyl benzyl-*N*-acetyl-D-galactopyranoside (Ac_3BnGalNAc, **1d**, substrate decoy inhibitor) (Agarwal *et al.*, 2013), peracetyl *N*-azidoacetyl-D-galactosamine (Ac_4GalNAz, **1e**, reporter analogue) (Hang *et al.*, 2003), peracetyl *N*-thioglycolyl-D-glucosamine (Ac_5GlcNTGc, **2a**, C-4 epimer of **1a**) (Wang *et al.*, 2021), and peracetyl *N*-acetyl-D-glucosamine (Ac_4GlcNAc, **2b**, C-4 epimer of **1b**) (Kim *et al.*, 2004) (**Fig. 1B**).

Inhibition of *O*-glycosylation decreases melanin biosynthesis in B16F10 cells

To study the role of *O*-glycosylation in melanogenesis, we employed B16F10 mouse melanoma cells, a well-established, highly pigmented melanoma model widely used to study melanogenesis and melanoma progression *in vitro* and *in vivo* (Fidler,

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1975). B16F10 cells were treated with a peracetylated GalNAc analogue: **1a**, **1b**, **1c**, or **1d**. Cells were incubated separately with each analogue (100 µM, 48 h), harvested, and intact cell suspensions were examined for visible melanin pigmentation. Strikingly, cells treated with **1a** showed no visible pigmentation compared to the other treatment groups (**Fig. 1C**). This depigmentation was specifically induced by **1a**, but not by **1b**, in a dose-dependent manner (25-200 µM) (**Fig. 1D**). Consistent with these observations, phase contrast microscopy of adherent cultures revealed abundant visible melanin granules in vehicle-treated cells, which were notably absent in **1a**-treated cells (**Fig. 1E**). Quantitative spectrophotometric analysis at 405 nm confirmed a ~ 75 % decrease in melanin content upon **1a** treatment compared to untreated controls. In contrast, treatment with DMSO (D, vehicle), **1b**, **1c**, and **1d** resulted in only modest decreases of 10 %, 25 %, 10 % and 35 %, respectively, compared to the untreated control (**Supporting Figure 1B**). The decrease in melanin content was dose-dependent (0-200 µM) for **1a**, whereas higher concentrations of **1b** and **1c** resulted in a slight increase in melanin content compared to vehicle treatment (**Fig. 1F**).

To assess the cytotoxicity and potential impact on cell proliferation, we evaluated the viability and growth of B16F10 cells following analogue treatment. MTT assays showed that **1a** and **1b** maintained 90 % cell viability at concentrations up to 100 µM, whereas **1c** caused moderate toxicity (~ 40 % decrease in viability) at 100 µM (**Supporting Figure 1C**). Trypan blue exclusion assay confirmed comparable viable cell numbers for D, **1b**, **1d**, and **1a**-treated cultures, with a ~ 20 % decrease observed only for **1c** (**Supporting Figure 1D and E**). Furthermore, carboxyfluorescein diacetate succinimidyl ester (CFSE) dilution analysis by flow cytometry revealed comparable proliferation profiles across all treatment groups after 48 h, indicating no cell cycle arrest in **1a**-treated cells (**Supporting Figure 1F**). Together, these results demonstrate that **1a** potently inhibits melanin biosynthesis in B16F10 cells, independently of cytotoxic or antiproliferative effects, and distinct from the effects of wild-type GalNAc (**1b**), the *N*-glycolyl analogue (**1c**), or the substrate decoy inhibitor (**1d**).

Disruption of Pmel17/gp100 glycosylation by 1a

Having confirmed the inhibition of melanin biosynthesis by **1a**, we examined its effects on the glycosylation of Pmel17/gp100, a melanosomal-specific structural protein. Based on our prior observations that **1a** induces hyposialylation of the heavily *O*-glycosylated mucin-domain protein CD43 (sialophorin), we anticipated that it would similarly perturb *O*-glycosylation and terminal sialylation of Pmel17/gp100 (Agarwal *et al.*, 2013). In mice, Pmel17/gp100/silv is a 626-amino acid multi-domain protein consisting of a signal peptide (SP), *N*-terminal region (NTR), polycystic kidney disease (PKD) domain, repeat (RPT) domain, cleavage site (CS), kringle-like domain (KLD), and *C*-terminal domain (CTD) (**Fig. 2A**).

Following synthesis, Pmel17/gp100 undergoes extensive post-translational glycosylation and proteolytic processing, generating multiple functional fragments, including the *N*-terminal Mα, Mα-Mβ, PKD-RPT, and *C*-terminal Mβ fragments (Berson *et al.*, 2003). The PKD-RPT domain is particularly critical for the formation of the organized, striated amyloid fibrillar architecture required for normal melanosome function. Although Pmel17/gp100 is known to be heavily glycosylated, the extent and functional significance of this glycosylation remain incompletely characterized.

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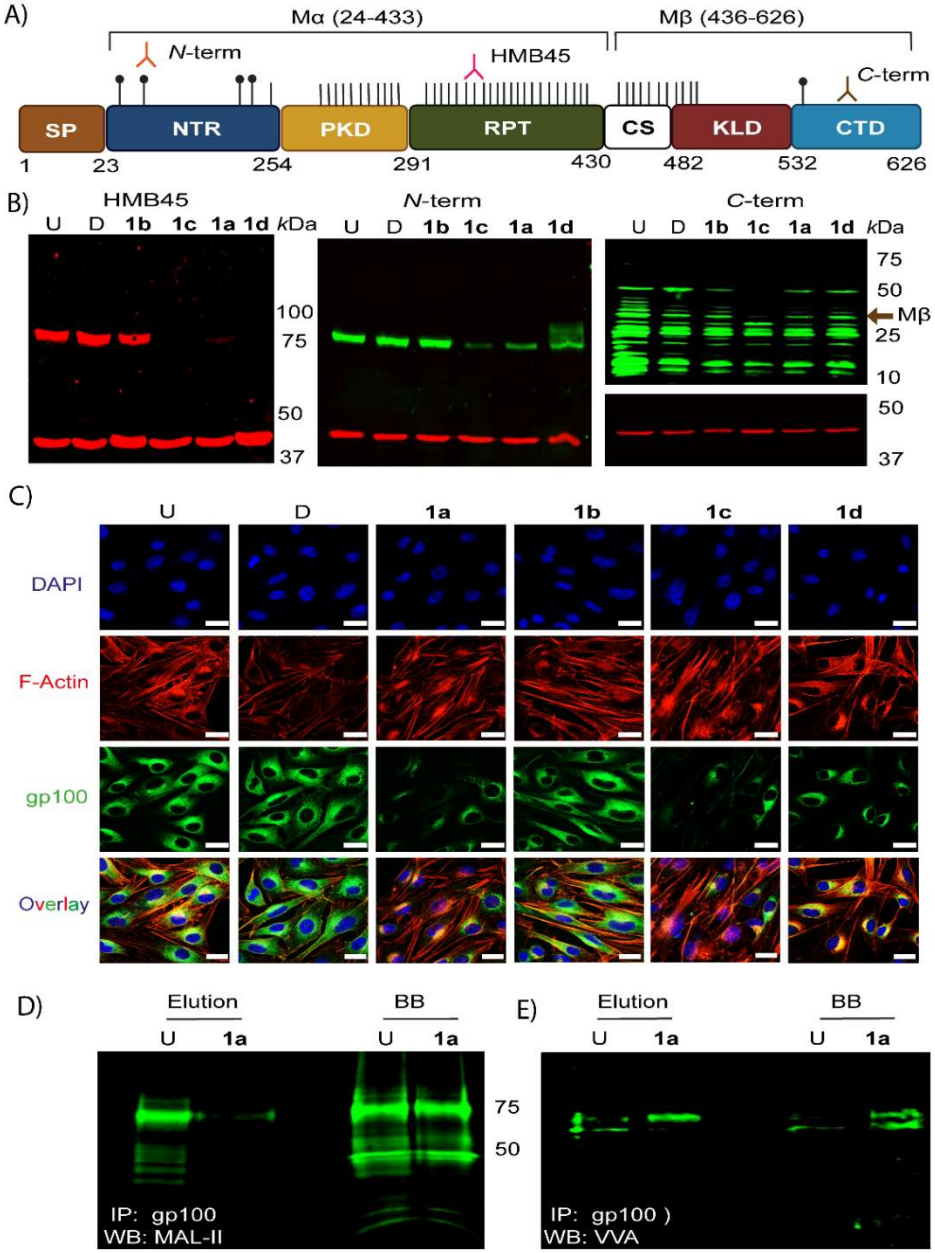


Figure 2. Modulation of Pmel17/gp100 glycoforms upon analogue treatment in B16F10 cells. (A) Schematic representation of predicted glycosylation sites on mouse Pmel17/gp100. The 626-amino acid protein is modified by four *N*-linked glycans and ~60 potential *O*-linked glycans, with 56 sites clustered within the PKD and RPT domains that contribute to fibril formation. SD, signal domain; NTD, *N*-terminal domain; PKD, polycystic kidney disease domain; RPT, repeat domain; CS, cleavage site; KRG, kringle-like domain; TM, transmembrane; and CTD, *C*-terminal domain. Amino acid positions marking the start of each domain are indicated below. *O*-glycosylation sites are indicated by black lines and *N*-glycosylation sites by lines with circles. Binding regions for glycan-dependent (HMB45), *N*-terminal, and glycan-independent *(C*-terminal) Pmel17/gp100 antibodies are indicated. (B) Duplex immunoblot analysis of Pmel17/gp100 in B16F10 cells treated with GalNAc analogues (100 μM, 48 h). Blots were probed with glycan-dependent HMB45 (left panel) (upper red band), N-term (middle panel) (upper green band), or glycan-independent C-term (right panel) (upper green band), with β-actin as a loading control (lower red band). Images were acquired on an Amersham Typhoon fluorescent imager. Blots shown are representative of three biological replicates (BR). (C) Confocal imaging of B16F10 cells treated with GalNAc analogues (100 μM, 48 h) and stained for Pmel17/gp100 (*N*-term, green), F-actin (phalloidin, red), and nuclei (DAPI, blue). Imaging was performed using 63× oil immersion on a Zeiss LSM 980 confocal microscope. The images shown are representative of three BRs, with two technical replicates per BR. Each replicate was conducted on a single cover slip, with five fields imaged per coverslip. Scale bar: 5 μm. (D-E) Lectin blot analysis of immunoprecipitated Pmel17/gp100 eluted by 0.2 M Glycine (pH 2.6) showing (D) reduced MAL II binding and (E) increased VVA binding upon **1a** treatment, indicating loss of terminal α2,3-linked sialic acids and exposure of the Tn antigen, respectively. Both the eluted fraction and the boiled beads (BB) fractions are shown. The data are representative of three BR. MAL II (*Maackia amurensis* Lectin II); VVA (*Vicia villosa* agglutinin); U, untreated; D, DMSO (vehicle); **1a**, Ac_5GalNTGc; **1b**, Ac_4GalNAc; **1c**, Ac_5GalNGc; and **1d**, Ac_3BnGalNAc.

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Notably, several gp100-derived peptides presented on MHC molecules have been explored as melanoma vaccine candidates, including $gp100_{25\text{-}33}$ (KVPRNQDWL) and $gp100_{44\text{-}59}$ (WNRQLYPEWTEAQRLD), which originate from NTR, as well as $gp100_{209\text{-}217}$ (ITDQVPFSV) and $gp100_{280\text{-}288}$ (YLEPGPVTA), derived from the PKD domain. Interestingly, these immunogenic epitopes predominantly arise from relatively non-glycosylated regions of Pmel17/gp100, highlighting a potential limitation of tumor antigen studies that do not consider the impact of glycosylation on antigen processing and presentation (Slingluff *et al.*, 2001; Schwartzentruber *et al.*, 2011; Chang and Xia, 2015).

Analysis of the Pmel17/gp100 amino acid sequence (Uniprot Entry No. Q60696) using NetNGlyc and NetOGlyc predicted four *N*-glycan sites and 60 *O*-glycan sites, respectively, predominantly clustered within the PKD and RPT domains (**Fig. 2A**) (Gupta and Brunak, 2002; Steentoft *et al.*, 2013). The dense clustering of *O*-glycans is proposed to create a bottle-brush-like structure essential for striated fibril formation and its stabilization. Hearing *et al.* have demonstrated that Pmel17/gp100, specifically the RPT domain, is decorated with core 1 *O*-glycans carrying Tn, sialyl-Tn, sialyl-T, and disialyl-T structures (Valencia *et al.*, 2007).

To assess **1a**-induced changes in Pmel17/gp100 glycosylation, we employed western blotting using a panel of three antibodies, *viz.*, (i) anti-gp100 (*N*-term), recognizing the *N*-terminal region; (ii) anti-gp100 (HMB45), which binds sialoglycans within the RPT domain of Pmel17/gp100, and (iii) anti-gp100 (*C*-term), recognizing the *C*-terminal region of Pmel17/gp100 (Valencia *et al.*, 2007). B16F10 cells were treated with the GalNAc analogues (100 μM, 48 h), harvested, lysed, resolved by SDS-PAGE, transferred to nitrocellulose membranes, and probed with anti-gp100 antibodies.

Strikingly, anti-gp100 (HMB-45) epitopes (sialylated glycans) were abrogated upon treatment with **1a**, **1c**, and **1d**, while D (vehicle) and **1b** showed no changes compared to the untreated control (**Fig. 2B**, left panel). Anti-gp100 (*N*-term) blots showed a significant decrease in the 75 *kDa* bands upon **1a** and **1c** treatment, while **1d** treatment produced diffuse higher molecular-weight bands above 75 *kDa*, and D and **1b** treatment resulted in no changes (**Fig. 2B**, middle panel). However, although **1c** altered Pmel17/gp100 glycoforms, this occurred under conditions associated with moderate cytotoxicity and did not translate into decreased melanin content.

In the anti-gp100 (*C*-term) blot, U, D, **1b**, and **1d** treatments showed no change in the 28 *k*Da band intensity, and importantly, **1a** also did not alter *C*-terminal protein levels, demonstrating that its effect was restricted to post-translational glycosylation without affecting overall Pmel17/gp100 expression or proteolytic processing (**Fig. 2B**, right panel). In contrast, **1c** treatment resulted in a clear decrease in the 28 *k*Da band, indicating decreased Pmel17/gp100 protein stability or enhanced degradation under these conditions. The effects of **1a** were found to be dose-dependent (0-200 μM, 48 h) and reversible. Suppression of Pmel17/gp100 glycosylation became evident at 50 μM and was maximal at 100-200 μM, as detected by both anti-gp100 (*N*-term and HMB45) antibodies (**Supporting Figure 2B and C**). Time-course analysis with a single dose of **1a** (100 μM, 0-96 h) showed maximal suppression at 48-72 h, with full recovery by 96 h, reflecting normal biosynthetic turnover of glycoproteins, analogue dilution, and endogenous GalNAc restoration (**Supporting Figure 2D**). Confocal microscopy corroborated these findings, revealing a near-complete loss of anti-gp100

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(*N*-term and HMB45) epitopes upon treatment with **1a**, **1c**, or **1d**, but not with D or **1b** (**Fig. 2C and Supporting Figure 2A**). F-actin staining using rhodamine-conjugated phalloidin served as an internal control, confirming that overall cellular morphology remained unaffected across all treatments.

To show that **1a** directly alters the Pmel17/gp100 glycosylation status, Pmel17/gp100 was immunoprecipitated from B16F10 cell lysates using anti-gp100 (*N*-term) antibody. Lectin blotting of immunoprecipitated Pmel17/gp100 revealed a marked reduction in MAL-II (*Maackia amurensis* lectin-II) binding upon **1a** treatment, compared to control, indicating a specific loss of terminal α2→3-linked sialic acids on the protein (**Fig. 2D**). Conversely, VVA (*Vicia villosa* agglutinin) binding to immunoprecipitated Pmel17/gp100 was increased, indicating exposure of the underlying Tn antigen (GalNAcα-Ser/Thr) (**Fig. 2E**). Notably, lectin blotting of the boiled bead fraction, which captures all antibody-bound Pmel17/gp100 including strongly bound or incompletely eluted species, exhibited the same trend as the eluted fraction, confirming that the observed glycosylation changes were robust and independent of elution efficiency.

These results provide direct molecular evidence that **1a** inhibits the elongation of *O*-glycans on Pmel17/gp100. Mechanistically, it is plausible that metabolic incorporation of **1a** perturbs the activity or substrate preference of glycopeptide-selective core 1 β1→3-galactosyltransferase (C1GalT1), thereby impairing the coordinated “filling-in” of adjacent *O*-glycosylation sites (Gill, Clausen and Bard, 2011; Wang *et al.*, 2021). Such disruption would decrease overall site occupancy and glycan density on Pmel17/gp100, resulting in immature *O*-glycan structures and protein hyposialylation.

Selective disruption of melanosomal glycoprotein enzymes by 1a while sparing structural proteins

To define the selectivity of **1a** for *O*-glycosylated proteins, we analyzed melanosomal proteins with contrasting glycosylation status. Tyrosinase is a critical *O*-glycosylated enzyme required for the initiation and regulation of melanin biosynthesis, and its inhibition underlies the mechanism of several anti-melanogenic agents (Mikami *et al.*, 2013). *In silico* analysis using Net-*N*-Glyc and Net-*O*-Glyc software predicted five *N*-glycan sites and four putative *O*-glycan sites in the mouse tyrosinase glycoprotein (Uniprot Entry No. P11344). Western blot analysis revealed a pronounced reduction in tyrosinase glycoprotein levels following **1a** treatment, but not upon treatment with D (vehicle), **1b** (wild-type control), or **1c** (*N*-glycolyl analogue), when compared to the untreated control (**Fig. 3A**). This is similar to the glycan-dependent loss of Pmel17/gp100 and strongly supports a requirement for intact glycosylation in maintaining tyrosinase stability and activity.

To validate the specificity of **1a** for *O*-glycosylated proteins, we examined MART-1/Melan-A, a non-glycosylated melanosomal structural protein that facilitates melanosome maturation and stabilizes Pmel17/gp100 (Hoashi *et al.*, 2005). In contrast to Pmel17/gp100, which harbors ~60 predicted *O*-glycan sites and four *N*-glycan sites, MART-1 is predicted to carry neither *O*- nor *N*-glycans. Western blotting and confocal microscopy with an anti-MART-1 antibody demonstrated that MART-1 expression

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remained unaffected by **1a** treatment at the 100 µM concentration used (**Fig. 3B** and **3C**). Immunoblotting revealed a distinct band at ~16 kDa in untreated, D-, **1b-, 1d-**, and **1a**-treated cells. This indicates that **1a** does not affect the levels of non-glycosylated melanosomal proteins. In contrast, **1c** treatment led to a decline in MART-1 expression, consistent with its recognized cellular toxicity at elevated levels rather than selective *O*-glycosylation inhibition.

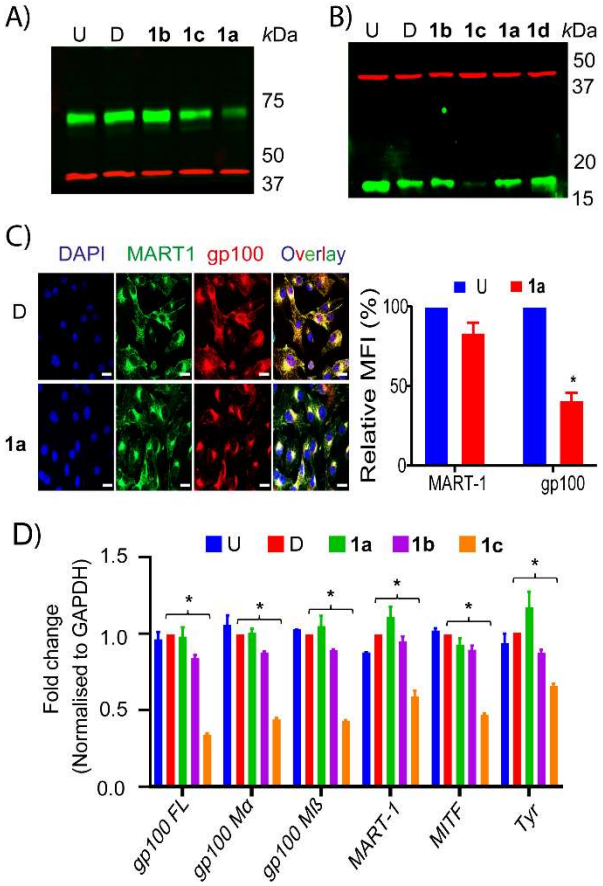


Figure 3. Selective post-translational effects of GalNAc analogues on melanosomal proteins. (A) Effect of GalNAc analogues on tyrosinase levels in B16F10 cells. Cells were treated with GalNAc analogues (100 µM, 48 h), lysed, and probed with anti-tyrosinase antibody (upper green bands) and anti-β-actin (lower red bands; loading control). (B) Effect of GalNAc treatment on MART-1 levels in B16F10 cells. Cells were treated with GalNAc analogues (100 µM, 48 h), lysed, and probed with anti-MART-1 antibody (upper green bands) and anti-β-actin (lower red bands; loading control). (C) Confocal imaging of B16F10 cells either left untreated or treated with **1a** (100 µM, 48 h), fixed, permeabilized, and stained for MART-1 antibody (green), Pmel17/gp100 (*N*-term, red), and nuclei (DAPI, blue). Imaging was performed using a Zeiss LSM 980 confocal microscope with a 63× objective. The merged images show colocalization of MART-1 with Pmel17/gp100 (yellow). The images shown are representative of three BRs, with two technical replicates per BR. Each replicate was conducted on a single coverslip, with five fields imaged per coverslip. Scale bar: 5 µm. (D) RT-qPCR analysis of expression of melanosomal-specific genes upon GalNAc analogue treatment in B16F10 cells. Transcript levels of Pmel17/gp100 (full-length, Mα, and Mβ), MART-1, MITF, and tyrosinase were quantified following analogue treatment (100 µM, 48 h). Error bars represent the standard deviation of three BR with three technical replicates for each BR. Statistical analysis was performed using two-way ANOVA. P values: *<0.05 compared to vehicle, D. U, untreated; D, DMSO (vehicle); **1a**, Ac_5GalNTGc; **1b**, Ac_4GalNAc; **1c**, Ac_5GalNGc; **1d**, Ac_3BnGalNAc; and FL, Full length.

RNA transcript levels were measured by qRT-PCR using primer sets targeting full-length Pmel17/gp100 (FL) as well as regions corresponding to the Mα and Mβ processing fragments to exclude the possibility that altered protein patterns arose from differential transcript expression or alternative splicing. Across all primer sets, **1a** treatment did not significantly change Pmel17/gp100 transcript levels compared to U and D controls (**Fig. 3D**). Similarly, transcripts of other key melanogenic regulators,

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including MART-1, MITF, and tyrosinase, remain unchanged upon **1a** treatment. Importantly, **1b** also showed transcript levels comparable to U and D controls, whereas **1c** exhibited a modest decrease in several targets, consistent with its partial cytotoxic effects (**Fig. 3D**). Together, these data demonstrate that **1a** does not affect transcription of melanosomal structure or regulatory genes.

Collectively, these data showed that **1a** acts with marked selectivity at the post-translational level by perturbing the glycosylation of melanosomal proteins; tyrosinase and Pmel17/gp100 are selectively inhibited, while non-glycosylated proteins, such as MART-1 remain unaffected. This selectivity confirms the direct mechanistic link between *O*-glycosylation inhibition and melanogenesis blockade.

1a induces hyposialylation, disrupts Siglec-E recognition, and arrests the migration and invasion of melanoma cells

Melanin is not only a defining feature of melanocytes and benign nevi but is also tightly linked to melanoma biology, where altered pigmentation states correlate with tumor progression, immune evasion, and metastatic potential (Slominski *et al.*, 2022). Having demonstrated that **1a** selectively disrupts *O*-glycosylated melanosomal proteins, including Pmel17/gp100 and tyrosinase, we next examined whether these changes extended to global cellular glycosylation. To address this, we employed a panel of lectins with distinct binding specificities: MAL-II, which binds NeuAcα2→3Gal moieties; *Sambucus nigra* agglutinin (SNA), which recognizes NeuAcα2→6Gal/GalNAc epitopes; wheat germ agglutinin (WGA), which binds both α2→3 and α2→6 sialylated residues; and VVA, which recognizes Tn-antigen (GalNAcα-Ser/Thr) (**Fig. 4A**) (Gupta, Surolia, and Sampathkumar, 2010).

B16F10 melanoma cells were incubated with GalNAc analogues (100 μM, 48 h), fixed, and stained with FITC-conjugated WGA to visualize cell-surface sialylation. Confocal microscopy revealed robust WGA staining on U, D, and **1b**-treated control cells, indicating abundant sialoglycans on the cell surface. In contrast, **1a**-treated cells displayed significantly decreased WGA staining, confirming a decrease in cell-surface sialoglycan abundance (**Fig. 4B**). Treatment with **1c** also resulted in decreased WGA staining, though the decrease was less pronounced than that observed with **1a**. In comparison, **1d** produced a modest decrease in WGA intensity.

Given the confocal microscopy findings, we examined global changes in glycosylation using lectin blotting. B16F10 cells were incubated with GalNAc analogues (100 μM, 48 h), lysed, resolved by SDS-PAGE, transferred to nitrocellulose membranes, and probed with biotinylated lectins. Levels of MAL-II epitopes (α2→3 sialic acids) were markedly decreased upon treatment with **1a** and **1c**, whereas **1d** showed minimal change relative to U, D (vehicle), and **1b** controls (**Fig. 4C**). Consequently, VVA binding (Tn-antigen) was significantly increased in **1a**-treated cells and was even more pronounced in **1d**-treated cells, while remaining largely unchanged in D, **1b**, or **1c** conditions (**Fig. 4D**). Together, these findings indicate that loss of terminal sialylation unmasks underlying Tn-antigen structures, resulting in increased VVA accessibility. Notably, whereas **1a** concomitantly decreased MAL-II binding and increased VVA signal, consistent with impaired core 1 elongation and reduced site occupancy. The substrate decoy **1d** induced robust Tn accumulation without a substantial decrease in MAL-II, suggesting a decoy-like effect that promotes

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accumulation of immature *O*-glycan structures without extensive removal of pre-existing α2→3 sialylated glycans.

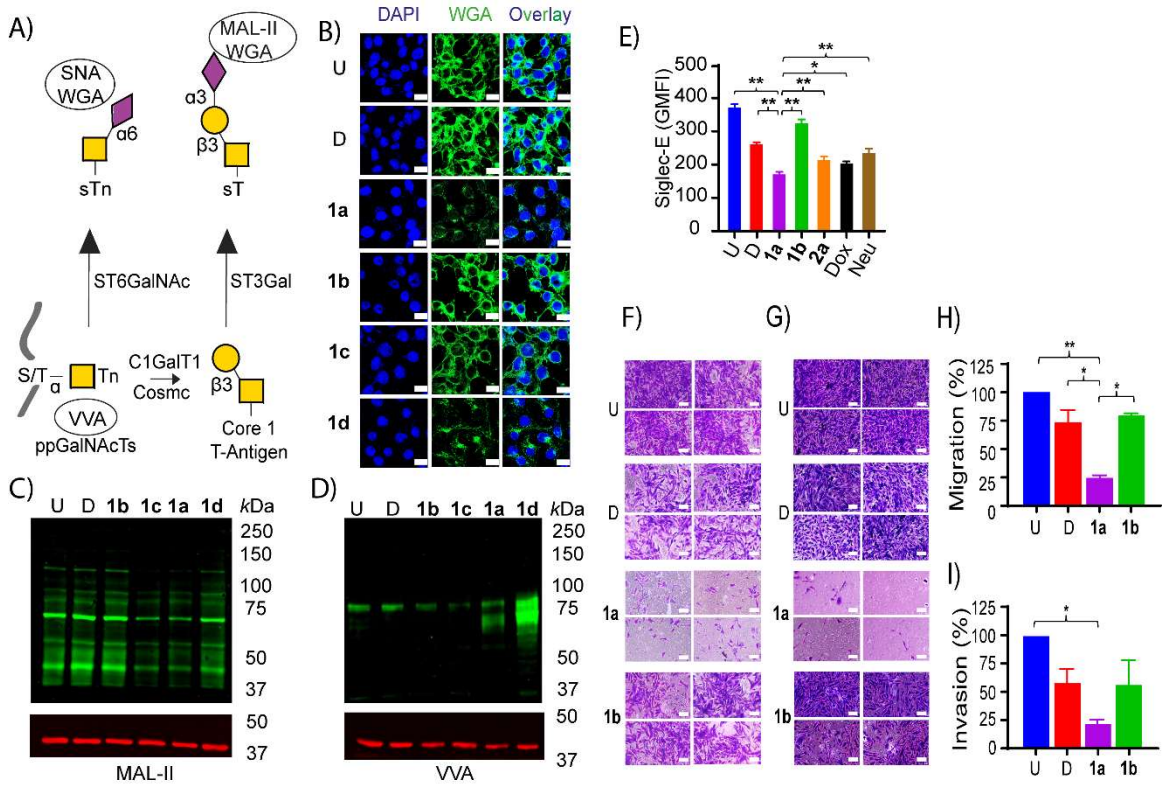


Figure 4. 1a causes hyposialylation, decreases Siglec-E binding, and suppresses melanoma migration and invasion. (A) Schematic representation of the binding specificity of lectins used in this study. (B) Confocal microscopy of B16F10 cells left untreated (U) or treated with D, **1b**, **1c**, **1a,** or **1d** (100 μM, 48 h), fixed, permeabilized, and stained with fluorescein-conjugated wheat germ agglutinin (WGA) (green) and DAPI (blue). Imaging was performed using a Zeiss LSM 980 confocal microscope with a 63× objective. The images shown are representative of three BRs, with two technical replicates per BR. Each replicate was performed on one cover slip, with five fields imaged per cover slip. Scale bar: 5 μm. (C-D) Lectin blot analysis of B16F10 cells treated with GalNAc analogues (100 μM, 48 h), lysed and probed with lectins: (C) *Maackia amurensis* lectin-II (MAL-II) (α2→3 sialic acids), and (D) *Vicia villosa* agglutinin (VVA) (Tn-antigen). β-actin blots are shown as loading controls. Blots are representative of three BR. (E) Flow cytometric analysis of Siglec-E binding to B16F10-Luc2 melanoma cells (1.0×10^6) left untreated (U) or treated with vehicle (DMSO), **1b** (100 μM), **1a** (100 μM), **2a** (100 μM), or Dox (100 nM) for 48 h. Cells treated with α2→3, 6, 8-neuraminidase from *Clostridium perfringens* (100 units/mL) for 10 min were used as additional controls. Cells were gated sequentially to exclude debris, doublets, and dead cells (**Supporting Figure 3G**). Siglec-E-Fc binding was quantified as geometric mean fluorescence (GMFI) of the Alexa Fluor 488(AF488)-positive cells. Error bars indicate the SD of three BR with two technical replicates each. Statistical significance was determined by one-way ANOVA with Tukey's multiple comparisons test. P-value * ≤ 0.05, and P-value ** ≤ 0.01. (F-I) Transwell migration and invasion assays of B16F10-Luc2 (1.0×10^5) cells left untreated (U) or treated with vehicle (DMSO), **1b** (100 μM), or **1a** (100 μM). Representative images of cells (F) migrated and (G) invaded to the basal side of the filters, fixed, and stained with 0.25 % crystal violet, are shown. [Scale bar: 100 μm, images shown are representative of four fields from one replicate, 10× objective]. Quantification of migration (H) and invasion (I) was performed by solubilizing the cells that migrated to the basal side in DMSO and measuring absorbance at 570 nm. Error bars represent the SD of three BR. Statistical significance was determined using one-way ANOVA with Tukey's multiple comparisons test. P-value * ≤ 0.05 and P-value ** ≤ 0.01. U, untreated; D, DMSO (vehicle); **1a**, Ac_5GalNTGc; **1b**, Ac_4GalNAc; **1c**, Ac_5GalNGc; **1d**, Ac_3BnGalNAc; **2a**, Ac_5GlcNTGc; Dox, Doxorubicin; and **Neu**, neuraminidase. Monosaccharide symbols follow the Symbol Nomenclature for Glycans (SNFG) (Neelamegham *et al.*, 2019).

To quantify surface-level glycosylation in intact cells, we performed flow cytometry. SNA binding decreased by approximately 50 % upon **1a** treatment compared to controls, confirming a decrease in α2→6-linked sialic acids on the cell surface. Correspondingly, VVA binding increased by 10 % upon **1a** treatment, indicating enhanced exposure of Tn-antigens (**Supporting Figure 3A, B, and C**). Importantly, neuraminidase treatment did not further increase VVA binding,

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demonstrating that the observed signal reflects genuine accumulation of Tn antigen (GalNAcα-Ser/Thr) due to impaired *O*-glycan elongation upon **1a** treatment, rather than secondary exposure of underlying structures following enzymatic removal of terminal sialic acids from mature *O*-glycans. Collectively, these findings establish that **1a** induced hyposialylation in melanoma cells by inhibiting mucin-type *O*-glycan biosynthesis, as reflected by decreased sialoglycans (WGA, SNA, and MAL-II) and increased Tn-antigen exposure (VVA).

Having established that **1a** treatment induces hyposialylation and increases Tn-antigen exposure, we investigated the functional consequences of decreased sialoglycans on immune recognition. Siglecs constitute a family of immune receptors that recognize sialylated glycans and activate immunosuppressive signals through ITIMS (Crocker, Paulson, and Varki, 2007). Among the five murine Siglecs, Siglec-E, expressed predominantly on macrophages and neutrophils, has been directly implicated in the immune evasion of B16F10 melanoma and tumor progression by engaging multi-valent tumor-associated sialoglycans (Ibarlucea-Benitez *et al.*, 2021). This interaction protects malignant cells from immune-mediated clearance, thereby promoting tumor growth and metastasis.

To assess whether **1a**-induced hyposialylation functionally disrupts the Siglec-sialoglycan interactions, we measured binding of recombinant mouse Siglec-E-Fc chimeric protein to B16F10-Luc2 cells. Cells were incubated with GalNAc analogues (100 μM, 48 h), harvested, stained with pre-complexed Siglec-E-Fc/anti-Fc antibodies, and analyzed by flow cytometry. Robust Siglec-E-Fc binding was observed in untreated cells compared to the isotype control, consistent with the high basal sialoglycan expression on melanoma cell surfaces. Treatment with vehicle (D) and **1b** (wild-type control) resulted in a modest decrease of 30 % and 15 %, respectively, which likely reflects non-specific effects of DMSO exposure and mild perturbations in cellular metabolism by metabolic processing of acetylated monosaccharides on overall glycan biosynthesis (**Fig. 4E and Supporting Figure 3D**). Strikingly, **1a** treatment resulted in a 50 % decrease in Siglec-E-Fc binding compared with the untreated control, demonstrating potent disruption of the Siglec-sialoglycan immune axis.

To assess whether this effect reflects a general consequence of hexosamine analogue-mediated glycosylation perturbation rather than selective inhibition of MTOG, we examined Ac_5GlcNTGc (**2a**), the C-4 epimer of **1a**, which is known to preferentially affect non-mucin glycans. Treatment with **2a** produced a moderate 40 % decrease in Siglec-E-Fc binding, suggesting that non-mucin sialoglycans may contribute to receptor engagement but are insufficient to account for the full inhibitory effect observed with **1a**. In contrast, **1a** elicited a more pronounced reduction, supporting a dominant role for mucin-type *O*-glycosylated ligands in mediating Siglec-E binding on melanoma cells. Doxorubicin (Dox), included as a cytotoxic chemotherapeutic control, resulted in a 45 % decrease, likely reflecting non-specific alterations in surface glycoproteins under cellular stress rather than selective glycosylation inhibition. Enzymatic desialylation with neuraminidase yielded a 30 % decrease, validating assay specificity and confirming the central role of sialic acids in Siglec-E recognition. These results establish that **1a** treatment functionally disrupts the Siglec-E-sialoglycan immune checkpoint by decreasing tumor cell surface sialylation.

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Having confirmed that **1a** profoundly disrupts *O*-glycosylation and decreases cell-surface sialoglycans, we investigated the functional consequences for key metastatic properties of melanoma cells, specifically migration and invasion. Migration was evaluated using a Boyden chamber assay. B16F10-Luc2 cells were seeded onto trans-well membranes (8.0 µm pore size) and treated with GalNAc analogues **1a** or **1b** (100 µM) under non-chemotactic conditions (no serum gradient) until a confluent monolayer was established. A serum gradient was then applied, and cells were re-treated with the analogues (100 µM, 48 h). Cells that migrated to the basal side of the membrane were fixed, stained with crystal violet, imaged, subsequently solubilized in DMSO buffer, and quantified spectrophotometrically. Cells treated with the vehicle (D) or **1b** (wild-type control) showed robust migratory capacity comparable to the untreated control (**Fig. 4F and H**). In striking contrast, **1a**-treatment resulted in a drastic 75 % decrease in migratory cells at 100 µM, with a similar inhibition evident at 50 and 200 µM concentrations (**Supporting Figure 3E**).

To evaluate the invasive capacity, we employed a matrigel-coated Boyden chamber invasion assay. B16F10-Luc2 cells treated with D or **1b** showed invasive behavior similar to that of the untreated control (**Fig. 4G and I**). However, **1a**-treatment resulted in a striking 75 % decrease in invasive capacity at 100 µM, with comparable inhibition observed at both 50 and 200 µM concentrations (**Supporting Figure 3F**). These effects were specific to **1a**, as **1b** treatment did not impair migration or invasion, ruling out non-specific effects due to peracetylation of HexNAc analogues.

Selective anti-metastatic transcriptional reprogramming by 1a with minimal off-target effects

To investigate the molecular basis of the observed anti-metastatic effects, we first employed a targeted RT-qPCR array profiling 84 metastasis-associated genes in B16F10 cells treated with vehicle (D) or **1a** (100 µM, 48 h). This focused approach revealed that the expression of the majority of genes (70 of 84) was unaffected, indicating the absence of broad transcriptional disruption. Notably, **1a** treatment significantly upregulated tumor suppressors Ephb2 (ephrin type-B receptor 2) (2.2-fold) and Itga7 (integrin subunit alpha 7) (2.0-fold), while simultaneously downregulating both key pro-metastatic regulators MMP10 (matrix metalloproteinase-10) and Mycl1 (V-myc myelocytomatosis viral oncogene homolog 1) by ~50 % (0.5-fold) (Nesbit, Tersak, and Prochownik, 1999; Yu *et al.*, 2011; Zhang *et al.*, 2014; Guan *et al.*, 2020) (**Supporting Figure 4A and B**). This distinct transcriptional signature aligns with the pronounced inhibition of migration and invasion and supports a functional link between *O*-glycosylation inhibition, disruption of Siglec-sialoglycan interactions, and suppression of metastatic phenotypes *in vitro.*

While the metastasis-focused array provided high sensitivity toward predefined invasion- and migration-associated pathways, it does not assess genome-wide transcriptional specificity or potential off-target effects. Therefore, to independently evaluate the global transcriptional impact of **1a**, we performed unbiased RNA-seq analysis on D- and **1a**-treated B16F10 cells under identical conditions (100 µM, 48 h). Using the criteria (adjusted P-value <0.05, $|\log_2 Fc|>1.5$), only 35 genes out of 16,359 protein-coding genes were differentially expressed, with 15 upregulated and 20 downregulated (**Supporting Figure 4C-J**). Importantly, none of the canonical

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melanocyte/melanosomal-associated genes, including Pmel17/gp100, tyrosinase, TYRP1, or MITF, met the predefined differential expression threshold ($|\log_2 Fc|>1.5$), confirming that **1a** does not perturb the core melanocytic transcriptional program.

The limited set of differentially expressed genes identified by RNA-seq showed modest enrichment for thiol-group-mediated redox homeostasis and glutathione (GSH/GSSG) metabolism pathways, including Hmox1, Nqo1, Gsta3, and Slc7a11, suggesting a mild adaptive response involving glutathione-dependent detoxification and thiol buffering rather than generalized cytotoxicity. Crucially, several metastasis-relevant genes detected by the targeted RT-qPCR array did not meet RNA-seq significance thresholds, highlighting the complementary strengths of these approaches. RNA-seq establishes global transcriptional selectivity and rules out widespread off-target effects, whereas the focused metastasis array sensitively detects biologically meaningful changes in low-abundance, functionally specialized metastatic regulators. Together, these data indicate that **1a** selectively modulates a small subset of metastasis-associated transcripts without broad transcriptional reprogramming, supporting a mechanism primarily driven by post-translational modulation of *O*-glycosylation.

1a delays primary tumor growth and improves survival in a syngeneic mouse melanoma model

Having established that **1a** inhibits cell surface sialylation, Siglec-sialoglycan interactions, Pmel17/gp100 glycosylation, melanin biosynthesis, and melanoma cell migration and invasion *in* vitro, we investigated the therapeutic efficacy of GalNAc analogues *in vivo* using the syngeneic B16F10 melanoma model in C57BL/6J mice (Fidler and Nicolson, 1976). We employed B16F10-Luc2 cells, which stably express firefly luciferase, enabling non-invasive, real-time monitoring of tumor growth and progression through bioluminescence imaging (BLI), upon intraperitoneal administration of D-luciferin (Deng, McLaughlin and Klinke, 2017). We tested whether the intraperitoneally administered GalNAc analogues are systemically bioavailable and reach the subcutaneous tumor. To address this, we exploited metabolic glycan engineering (MGE) using Ac_4GalNAz (**1e**), a peracetylated azide-bearing GalNAc analogue, followed by bio-orthogonal ligation with DBCO-Cy5 through strain-promoted azide-alkyne cycloaddition (SPAAC, also known as click chemistry) (Laughlin *et al.*, 2006). After optimizing click reaction conditions *in vitro* (**Fig. 5A**), C57BL/6J mice were implanted with B16F10-Luc2 cells subcutaneously (1.0×10^6 cells per mouse, 50 µL) on day 1 and administered with either vehicle (DMSO, D) or **1e** (300 mg/kg, i.p.) on days 1, 3, 5, 7, and 9. On day 10, mice were euthanized, tumor masses were excised, and single-cell suspensions were prepared. Tumor cells were subjected to SPAAC with DBCO-Cy5, lysed, and resolved on an SDS-PAGE gel. Robust Cy5 fluorescence was detected exclusively in **1e**-treated samples, whereas vehicle-treated mice showed only a faint signal; silver-stained gels were used as loading controls (**Fig. 5B**). These results showed that intraperitoneally administered GalNAc analogues are systemically bioavailable and efficiently incorporated into tumor cell glycoproteins *in vivo*, likely facilitated by the enhanced permeability and retention (EPR) effect characteristic of solid tumors (Shinde *et al.*, 2022).

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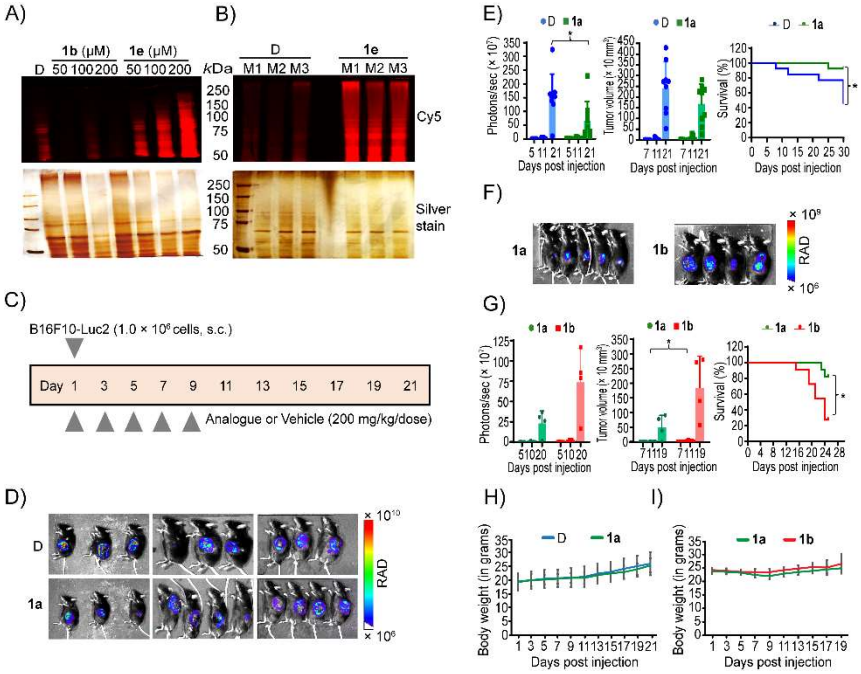


Figure 5. Tumor bioavailable-1a suppresses tumor growth and improves survival in a syngeneic melanoma mouse model. (A-B) Metabolic labeling confirms tumor bioavailability of intraperitoneally administered azido analogue (**1e**). (A) B16F10-Luc2 cells (1.0 × 10^6) were treated with **1b** or **1e** at 50, 100, or 200 μM for 48 h, followed by reaction with DBCO-Cy5 (10 μM, 30 min) via copper-free SPAAC. Cells were lysed using chloroform:methanol:water mixture, and proteins were precipitated with methanol. Protein samples (20 μL in SDS buffer) were resolved on 7.5% SDS-PAGE gels and imaged using a fluorescence scanner under the Cy5 (red) channel. Silver-stained gels are shown as loading controls. (B) *In vivo* bioavailability of **1e** at the tumor site. B16F10-Luc2 cells (1.0 × 10^6 cells in 50 μL) were injected subcutaneously into the left flank of C57BL/6J male mice (6 weeks old) on day one, followed by i.p. (intraperitoneal) administration of **1e** (300 mg/kg) or D once on days one, three, five, seven, and nine. On day 10, tumors were excised, homogenized, and single cell suspensions were clicked with 10 μM DBCO-Cy5 for 30 min. Proteins were processed as described in (A) and imaged under the Cy5 channel (n = 3 mice per group). Silver-stained gels are shown as loading controls. (C) Schematic representation of the experimental timeline and treatment regimen. (D) B16F10-Luc2 cells (1.0 × 10^6 cells in 50 μL) were injected subcutaneously into the left flank of C57BL/6J male mice under anesthesia on day one, followed by i.p. administration of **1a** (200 mg/kg) or D once on days one, three, five, seven, and nine. Tumor progression was monitored by bioluminescence imaging (BLI) after i.p. injection of D-luciferin; a representative image from day 21 is shown. Two mice in the vehicle-treated group died on day 4 due to poor health unrelated to treatment. (E) Quantification of BLI signal (photons/sec) and tumor volume (mm^3) measured using Aura software and vernier calipers, respectively, along with Kaplan-Meier survival analysis with an endpoint at day 30. Error bars represent the SD of n = 9 mice (three mice each in three independent experiments) for BLI/tumor volume and n = 13 for the survival curve. (F) B16F10-Luc2 cells (1.0 × 10^6 cells in 50 μL) were injected subcutaneously on day one, followed by 200 mg/kg i.p. doses of **1a** or **1b** once, on days one, three, five, seven, and nine. One mouse in the **1b**-treated group died on day 18 due to tumor burden. Error bars represent the standard deviation of n = 5 for BLI/tumor volume and n = 11 for the survival curve. (G) Corresponding BLI (photons/sec), tumor volume quantification (mm^3), and Kaplan-Meier survival curves with endpoints at day 25. Statistical significance was determined using multiple t-tests with the Holm-Sidak method (BLI and tumor volume) and the log-rank Mantel-Cox test (survival curve). P-value * ≤ 0.1 (BLI) and P-value * ≤ 0.05 (survival curve) in (E); P-value * ≤ 0.1 (tumor volume) and P-value * ≤ 0.01 (survival) in (G). (H-I) Body weight monitoring during treatment. Body weight was measured every alternate day. (H) Comparison between vehicle (DMSO) and **1a**-treated groups; (I) Comparison between **1a**- and **1b**-treated groups. D, DMSO (vehicle); **1a**, Ac_5GalNTGc; **1b**, Ac_4GalNAc; and **1e,** Ac_4GalNAz.

To evaluate therapeutic efficacy, B16F10-Luc2 cells (1.0 × 10^6 cells) were implanted subcutaneously on day 1, followed by intraperitoneal administration of vehicle (D, 100 μL) or **1a** (200 mg/kg, 100 μL) on days 1, 3, 5, 7, and 9 (five injections in total) (**Fig. 5C**). This study was performed across three independent cohorts with n = 3 mice per group per cohort (total n = 9 mice per group) to minimize inter-experimental variability in luciferase signal quantification arising from differences in luciferin administration and imaging conditions. Body weight was monitored on alternate days, tumor volume was measured on days 7, 11, and 21, and BLI was performed on days 5, 11, and 21. On day 21, representative BLI images revealed markedly decreased luminescence intensity in **1a**-treated mice compared with vehicle (D) control (**Fig. 5D**). Quantitative analysis showed an approximately 60 % decrease

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in photon emission (**Fig. 5E**, left panel) and a ~40 % decrease in tumor volume compared to the vehicle control which is taken as 100 % (**Fig. 5E**, middle panel), with no significant differences in body weight observed over the 21-day study period (**Fig. 5H**). Kaplan-Meier survival analysis revealed 100 % survival (n = 13) in the **1a**-treated group through day 21, compared to 70 % survival in the vehicle group, attributable to the reduced tumor burden in the **1a**-treated mice (**Fig. 5E**, right panel). The study was terminated on day 21 due to mortality observed in the vehicle-treated group. Remarkably, the tumor growth inhibition by **1a** persisted for 12 days after the final bolus administration on day 9, demonstrating durable therapeutic efficacy.

To validate the structure-activity relationships and confirm that the observed effects are specifically attributable to the thioglycolyl modification of **1a**, we compared **1a** with **1b** (wild-type control). Two groups (n = 5) of mice received B16F10-Luc2 cell implantation on day 1, followed by five intraperitoneal injections of either **1a** or **1b** on days 1, 3, 5, 7, and 9. On day 20, BLI images revealed a striking decrease in luminescence in the **1a**-treated cohort compared with **1b**-treated mice (**Fig. 5F**). **1a**-treated mice exhibited a significant 60 % lower tumor burden, as measured by both photoluminescence (day 20; **Fig. 5G**, left panel) and tumor volume (day 19; **Fig. 5G**, middle panel), with no differences in body weight between groups (**Fig. 5I**). Kaplan-Meier survival analysis again showed 100 % survival (n = 11) in the **1a**-treated group till day 21, compared to ~70 % survival in the **1b**-treated cohort (**Fig. 5G**, right panel).

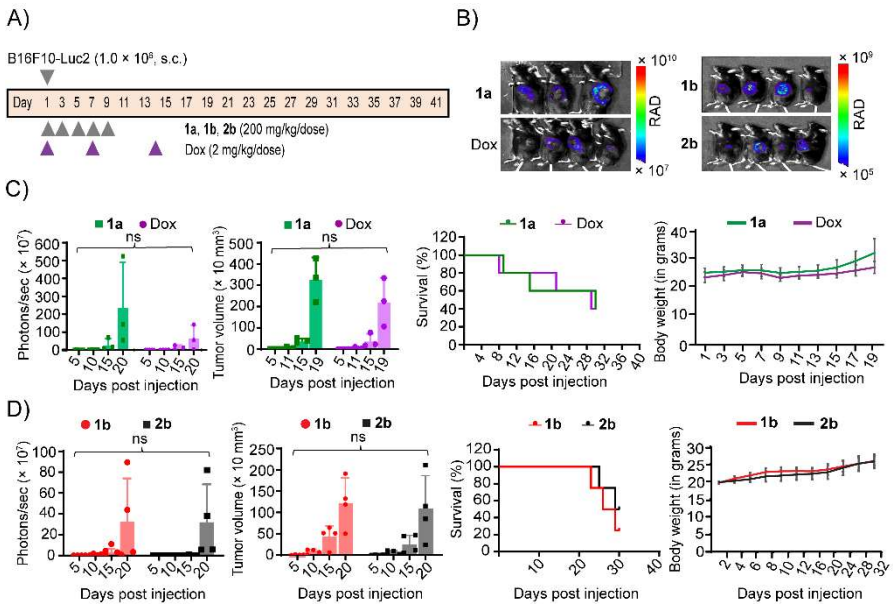


Figure 6. Benchmarking of 1a, and testing *in vivo* efficacy of 1b and its epimer 2b. (A) Schematic representation of the experimental timeline and treatment regimen. (B) Benchmarking of **1a** against Doxorubicin (Dox). B16F10-Luc2 cells (1.0×10^6 cells in 50 µL) were injected subcutaneously into C57BL/6J male mice on day one. This was followed by i.p. administration of either **1a** (200 mg/kg) once on days one, three, five, seven, and nine, or Dox (2.0 mg/kg) once on days one, seven, and 14. Comparison between **1b** and its C-4 epimer **2b**. Mice bearing B16F10-Luc2 tumors were tested intraperitoneally with **1b** or **2b** (200 mg/kg) once on days one, three, five, seven, and nine. Representative images from day 20 are shown. (C-D) BLI was captured on days five, 10, 15, and 20 after D-luciferin administration. One mouse in the **1a**-treated group died on day 9 and day 15 each, and one mouse in the Dox-treated group died on day 9. Tumor burden (photons/sec), tumor volume (mm^3), and body weight (grams) were quantified. Kaplan-Meier survival analysis was performed with an endpoint at day 30. Error bars represent the SD of n = 5 mice in all cases in (C) and n = 4 mice in (D). Statistical analysis was determined using multiple t-tests with the Holm-Sidak method (BLI and tumor volume) and the log-rank Mantel-Cox test (survival curve). All comparisons were non-significant. **1a**, Ac_5GalNTGc; **1b**, Ac_4GalNAc; and **2b**, Ac_4GlcNAc.

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Next, we benchmarked the efficacy of **1a** against doxorubicin (Dox), a standard anti-cancer agent (Carvalho *et al.*, 2009). Dox was administered at a dose of 2.0 mg/kg (once weekly, for three weeks), a regimen reported to decrease tumor growth while minimizing systemic toxicity (Mittal, Tabasum and Singh, 2014). Although Dox-treated mice showed greater tumor regression than the **1a** group, they also exhibited pronounced adverse effects, including hair loss and decreased body weight, which were notably absent in **1a**-treated mice. Importantly, overall survival rates were comparable in both **1a** and Dox-treated groups, indicating that **1a** achieves therapeutic benefit similar to standard chemotherapy but with a superior safety profile (**Fig. 6A-C)**. Finally, to confirm that the effects are strictly GalNAc-specific, we compared Ac_5GalNAc (**1b**) with Ac_5GlcNAc (**2b**), intraperitoneally. No significant differences in tumor volume were observed between mice treated with **1b** or **2b** (**Fig. 6A, B, and D**). Together with the lack of efficacy of non-thiolated (**1b**) or non-GalNAc analogues (**2b**), these data confirm that the antitumor activity of **1a** is structure-dependent and requires both GalNAc-mediated engagement and the thioglycolyl modification. This activity is likely mediated through selective modulation of mucin-type *O*-glycan biosynthesis and tumor-associated sialoglycan presentation on melanoma cells.

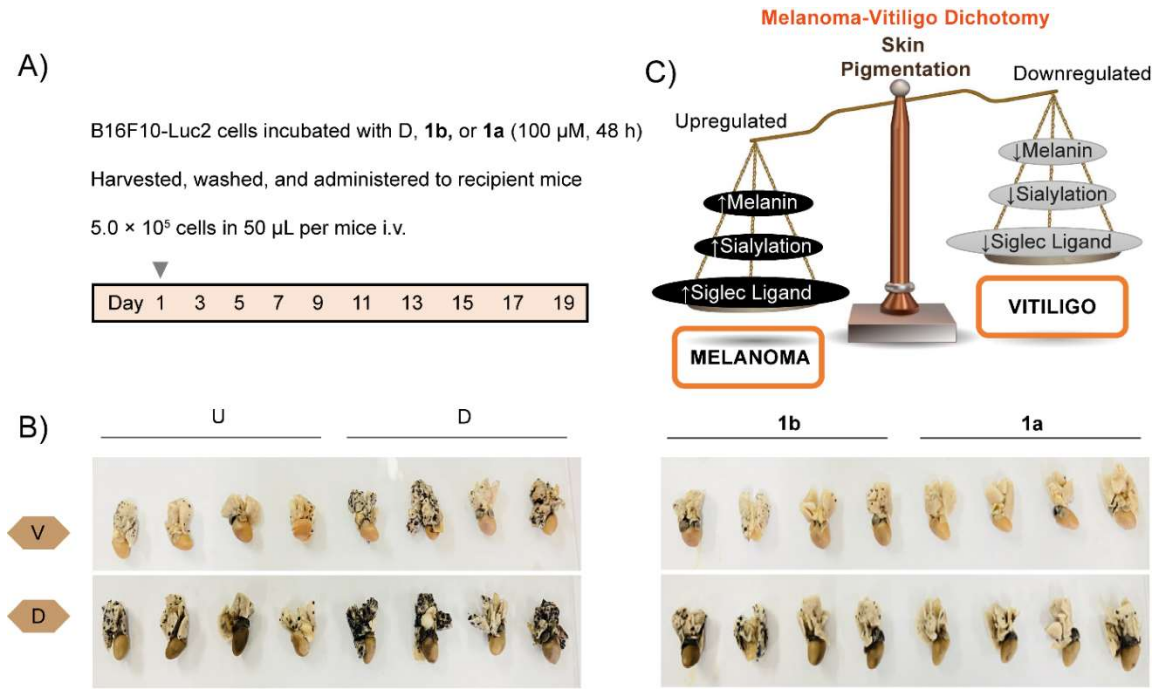


Figure 7. Intravenous injection of 1a-pretreated melanoma cells impairs metastatic lung colonization. (A) Schematic diagram showing the experimental timeline and treatment design. B16F10-Luc2 cells (5.0×10^5) were either untreated (U) or treated with D, **1a**, or **1b** (100 µM each, 48 h). Cells were harvested by trypsinization, washed thoroughly, resuspended in PBS, and 5.0×10^5 cells in 50 µL PBS were administered per mouse via the retro-orbital intravenous injection into C57BL/6J male mice (6 weeks old) under anesthesia. (B) Representative images of lungs (harvested together with the heart) collected on day 19 post-injection and imaged from both ventral (V) and dorsal (D) surfaces, showing metastatic melanoma burden. (C) Conceptual model illustrating the role of MTOG in regulating melanocyte and melanoma cell states. Balanced *O*-glycosylation supports normal pigmentation, whereas excessive melanin production and aberrant glycosylation contribute to melanoma progression and metastatic competence. Conversely, strong suppression of melanin synthesis is associated with hypopigmented states such as vitiligo and albinism. In the context of this study, **1a**-mediated inhibition of MTOG selectively decreases melanoma cells' surface sialylation, associated Siglec-ligand presentation, and perturbs Pmel17/gp100 glycosylation, thereby shifting melanoma cells away from a highly pigmented, immune-evasive, and metastatic phenotype without broadly altering gene expression programs. U, untreated; D, DMSO (vehicle); **1a**, Ac_5GalNTGc; and **1b**, Ac_4GalNAc.

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1a pre-treated melanoma cells show severely impaired lung colonization in an experimental metastasis model

Having established that **1a** treatment profoundly inhibits primary tumor growth and improves survival in orthotopic syngeneic melanoma models, we investigated whether MTOG inhibition has any effect on the metastatic potential of melanoma cells *in vivo*. To directly assess the impact of **1a** on tumor cell extravasation and metastatic colonization, we employed an experimental lung metastases model (Hart and Fidler, 1980). B16F10-Luc2 cells were pre-treated *in vitro* with vehicle (D, DMSO), wild-type control (**1b**), or **1a** (100 µM, 48 h), harvested, washed, and resuspended as single cell suspensions. Cells were then administered to C57BL/6J mice *via* retro-orbital intravenous injection (5.0×10^5 cells in 50 µL per mouse) (**Fig. 7A**). On day 19, mice were euthanized, and lungs were harvested.

Gross examination of lungs revealed striking differences in metastatic burden among treatment groups. Control mice receiving either vehicle-treated or **1b**-treated B16F10-Luc2 cells developed numerous dark, pigmented melanoma nodules on both dorsal and ventral lung surfaces, consistent with robust pulmonary colonization by melanoma cells (**Fig. 7B**). In sharp contrast, mice receiving **1a**-pre-treated melanoma cells showed a dramatic decrease in lung metastasis, with approximately 80 % fewer numbers of metastatic nodules compared to **1b**-treated controls (**Supporting Fig. 5**). Importantly, this profound suppression of lung colonization occurred despite intravenous injection of identical numbers of tumor cells, indicating that **1a** treatment intrinsically compromises metastatic competence rather than cell viability. These findings establish that cell-surface sialylation and mucin-type *O*-glycan presentation are critical for melanoma cell survival in circulation, endothelial extravasation, and secondary site colonization.

In the context of the proposed working model, **1a**-mediated inhibition of MTOG decreases melanoma cell surface sialylation and associated Siglec-ligand presentation, thereby weakening glycan-dependent immune inhibitory interactions and compromising metastatic fitness (**Fig. 7C and 1A**). Concomitantly, decreased sialylation may enhance peptide-MHC (pMHC) surface presentation or stability on tumor cells, potentially facilitating more efficient engagement of cognate T cell receptors. This effect would be consistent with mechanisms in which increased pMHC density promotes enhanced TCR micro-clustering and signaling (Singha *et al.*, 2017). Concomitantly, disruption of *O*-glycosylation perturbs Pmel17/gp100 maturation and melanosomal processing, leading to decreased melanin content and altered pigmentation states. Conceptually, this model places melanoma along a pigmentation and glycosylation continuum, wherein hyperpigmented, highly sialylated cells exhibit enhanced immune evasion and metastatic competence, whereas reduced melanin synthesis and attenuated sialoglycan display shift cells toward less aggressive phenotypes, analogous to hypopigmented conditions such as vitiligo. Importantly, these effects arise without broad reprogramming of tumor-intrinsic gene expression, highlighting the central role of post-translational glycosylation in controlling melanoma progression.

Collectively, these findings identify *O*-glycosylation-dependent cell-surface glycans as key regulators of metastatic dissemination and support pharmacological inhibition of MTOG as a promising neoadjuvant and adjuvant anticancer therapeutic

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strategy that targets the glyco-immune interface, rather than transcriptional oncogenic programs or kinase inhibitions.

Discussion

This study establishes mucin-type *O*-glycosylation (MTOG) as a critical determinant in melanoma progression and identifies it as a druggable therapeutic target. By employing a systematic structure-activity relationship approach with multiple GalNAc analogues, we identified Ac_5GalNTGc (**1a**) as a lead compound that selectively inhibits *O*-glycan biosynthesis through metabolic incorporation. Our *in vitro* studies unequivocally demonstrate that **1a** disrupts *O*-glycosylation through a mechanism distinct from halogen-substituted analogues that act *via* feedback inhibition of glycosyltransferases (Costa *et al.*, 2025). This mechanistic distinction is evidenced by the profound, dose-dependent decrease in melanin content (75 % decrease at 100 µM) and marked cell-surface hyposialylation, characterized by decreased MAL-II binding and increased VVA binding, thereby validating the thioglycolyl modification as a critical determinant of biological efficacy.

A key mechanistic insight from this work is that **1a** selectively disrupts Pmel17/gp100 glycosylation while sparing non-glycosylated melanosomal proteins such as MART-1, leaving the upstream transcriptional program intact, as confirmed by RNA-seq analysis. These findings establish that **1a** acts predominantly at the post-translational level rather than through transcriptional repression or generalized cytotoxicity. The selective loss of glycan-dependent epitopes recognized by HMB45 indicates that *O*-glycans play a structural role in maintaining melanosomal architecture. *O*-glycosylation likely stabilizes the organized, amyloid fibrillar sheets by limiting aberrant protein-protein interactions and maintaining appropriate spacing between fibrils, thereby preserving matrix organization and structural integrity. In melanoma cells, where melanosome integrity is tightly coupled to pigmentation, immune suppression, and metastatic competence, **1a**-induced hypoglycosylation generates structurally defective melanosomes. This results in an effective blockade of melanogenesis and attenuation of malignant phenotypes.

Notably, while conventional depigmenting agents predominantly target tyrosinase enzymatic activity, our findings identify upstream glycosylation as a powerful and previously underexplored regulatory layer of melanogenesis (Zolghadri *et al.*, 2019). This observation raises the intriguing possibility that age-related, inflammatory, or metabolic dysregulation of glycosylation pathways may contribute to pigmentary disorders such as vitiligo and lentigo (Sharma *et al.*, 2025). In these contexts, aberrant glycan loss or remodeling could destabilize melanosomes, decrease melanin synthesis, and expose cryptic antigenic determinants, thereby precipitating immune-mediated melanocyte destruction in vitiligo.

The molecular mechanisms underlying melanoma progression are intricately connected to melanogenesis, melanin content, and pigmentation dynamics, encompassing both hyperpigmentation, such as lentigo, and depigmentation disorders, including vitiligo (representing two sides of the same coin) (Wankowicz-Kalinska *et al.*, 2003). Conceptually, melanocyte biology can be viewed along a pigmentation continuum in which tightly regulated melanosome maturation and

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melanin production maintain normal skin pigmentation, whereas sustained, dysregulated melanogenesis supports melanoma progression, and impaired melanosome integrity or pigment synthesis leads to hypo-pigmentary disorders such as vitiligo (**Fig. 7C**) (Slominski, Paus and Mihm, 1998). Importantly, this continuum reflects functional melanocyte output rather than changes in melanocytic gene expression per se. By selectively perturbing melanosomal *O*-glycosylation without altering transcriptional programs, **1a** shifts melanoma cells toward a hypo-glycosylated, hypopigmented, and melanosome-defective state. This positions **1a**-treated cells as a valuable experimental model for interrogating glycosylation-dependent mechanisms governing melanosome stability, pigmentation, and immune recognition.

Our findings align with the growing evidence that melanogenesis and intact melanosome biology are closely linked to melanoma invasiveness and immune evasion (Lin and Fisher, 2007; Li *et al.*, 2022). Increasing evidence indicates that enhanced melanogenesis is associated with melanoma progression and immune modulation (Cabaço *et al.*, 2022). While melanin is photoprotective under physiological conditions, elevated melanin production in melanoma can promote oxidative DNA damage and support tumor evolution. Moreover, secreted melanin has been shown to remodel the tumor microenvironment by inducing cancer-associated fibroblasts, suppressing antitumor immunity, and promoting angiogenesis. Thus, increased melanogenesis may actively sustain melanoma aggressiveness rather than merely reflect differentiation status. In this context, selective attenuation of melanin synthesis through disruption of melanosomal *O*-glycosylation provides a mechanistically distinct strategy to impair both tumor-intrinsic and microenvironmental drivers of metastasis.

Treatment with **1a** markedly impaired melanoma cell migration and invasion *in vitro* and decreased Siglec-E binding by ~50 %, thereby functionally disrupting the immunosuppressive Siglec-sialoglycan axis. Importantly, these molecular and cellular effects translated into robust therapeutic efficacy *in vivo.* Using a syngeneic mouse melanoma model and the widely established B16F10 system expressing firefly luciferase (Luc2) that enables longitudinal tumor monitoring through BLI, **1a** significantly delayed primary tumor growth upon repeated administration. This antitumor activity was likely facilitated by enhanced tumor accumulation of **1a**, consistent with the enhanced permeability and retention (EPR) effect, whereby leaky tumor vasculature and impaired lymphatic drainage promote preferential retention of small molecules within the tumor microenvironment (Matsumura and Maeda, 1986). Tumor bioavailability of intraperitoneally administered **1a** was experimentally validated using the azido-containing GalNAc analogue **1e** as a metabolic probe, confirming its incorporation within tumor tissues. Upon further evaluation, **1a** profoundly impaired lung colonization in an experimental metastasis setting and significantly improved overall survival. Notably, these antitumor effects were comparable to those achieved with doxorubicin, yet occurred without detectable systemic toxicity or weight loss, highlighting the favourable therapeutic index associated with of targeting MTOG.

These results position *O*-glycan-directed strategies as an exciting therapeutic avenue in melanoma. Small molecules that disrupt the *O*-glycan biosynthesis have demonstrated antitumor activity in multiple tumor models. For example, analogue **1d** inhibits glycosyltransferase activity in human colon cancer cells, thereby suppressing tumor-associated glycan elaboration. While the sialic acid mimetic P-3Fax-Neu5Ac is

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metabolized to the CMP-derivative that broadly inhibits sialyltransferases, leading to a global decrease in sialylated epitopes and impairing tumor cell adhesion and migration (Huang *et al.*, 1992; Büll *et al.*, 2013). Notably, P-3Fax-Neu5Ac has been shown to cause nephrotoxicity (Macauley *et al.*, 2014).

Conclusion

In conclusion, this study demonstrates that pharmacological inhibition of MTOG, **1a**, exerts a dual anti-tumorigenic effect: (i) dismantling the melanosomal architecture essential for pigmentation, and (ii) disrupting Siglec-sialoglycan checkpoint-mediated immune evasion pathways. By impairing Pmel17/gp100 glycosylation and decreasing cell-surface sialylation, **1a** not only destabilizes melanosome structure but also diminishes glycan-dependent shielding of tumor-associated antigens. This decrease in sialylated ligands weakens inhibitory Siglec engagement and may enhance immune surveillance by permitting more effective recognition of tumor-associated peptide-MHC complexes. By acting at the level of post-translational glycan remodelling rather than gene expression, **1a** reveals glycosylation as a mechanistically distinct and therapeutically actionable vulnerability in melanoma. These findings highlight the largely untapped therapeutic potential of glycoengineering-based strategies in cancer treatment. Although further medicinal chemistry optimization and pharmacokinetic refinement will be required, **1a** represents a first-in-class MTOG-targeting agent that could be strategically deployed as a lead scaffold in combination with kinase inhibitors or immune checkpoint blockade to overcome therapeutic resistance and improve clinical outcomes in advanced melanoma (Larkin, Hodi and Wolchok, 2015).

Materials and Methods

Chemical Synthesis and Reagents

Peracetylated non-natural GalNAc analogues **1a**, **1b, 1c, 1e, 2a,** and **2b** were synthesized according to previously described protocols (Hang *et al.*, 2003; Du *et al.*, 2011; Agarwal *et al.*, 2013).

Cell Culture

B16F10 cells were cultured in sterile T-75 flasks or 100 mm tissue culture-treated dishes in DMEM supplemented with 10 % fetal bovine serum (FBS) and 1.0 % pen-strep (final concentration of 100 U/mL of penicillin and 0.1 mg/mL of streptomycin). Cultures were maintained in a humidified incubator at 37 °C with 5.0 % CO_2. For B16F10-Luc2 cells (a kind gift from Dr. Santiswarup Singha, NBL, NII), which constitutively express luciferase, the selection antibiotic Blasticidin S hydrochloride was added to a final concentration of 10 µg/mL. Cell density was determined using a particle counter (Beckman-Coulter Z2) by suspending 100 µL of cell suspension in 10 mL PBS. Only cells in the logarithmic growth phase were used for experiments. Cells were detached using 0.25 % trypsin-EDTA, quenched with ~10 mL of spent media, and pelleted by centrifugation (500 × g, 5.0 min). The supernatant was aspirated, and the pellet was resuspended in PBS and washed once by centrifugation (10 mL, 500 × g, 5.0 min). Hereafter, resuspending the cell pellet in PBS, ASB (avidin staining buffer),

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or LSB (lectin staining buffer), followed by centrifugation (500 × g, 5.0 min) to remove residual media, serum, antibodies, dye, or debris, is referred to as 'one wash cycle'.

Experiments *in vivo*

All animal protocols and experiments were approved by the Institutional Animal Ethics Committee (IAEC# 603/22) of the National Institute of Immunology (NII). Male C57BL/6J (WT) mice (4-6 weeks old) were procured from the Experimental Animal Facility (EAF), NII. Mice were provided food and water *ad libitum* and maintained under standard pathogen-free conditions. Parameters of weight, health, behavior, and morbidity were monitored regularly. Mice showing adverse parameters were excluded from the study. Animals were anesthetized via intraperitoneal (i.p.) administration of 100 µL of ketamine (80 mg/kg) and xylazine (20 mg/kg) solution before cell implantation, imaging, or analogue injection. The anesthetic effect lasted for approximately 20-30 min, which was sufficient for all procedures. Animals were euthanized using CO_2 asphyxiation whenever applicable. For i.p. injections, stock solutions of monosaccharide analogues and doxorubicin were prepared in DMSO and sterile filtered through 0.2 µm PTFE syringe filters. 100 µL of sterile-filtered DMSO served as the vehicle control.

Estimation of Melanin Content

Melanin content was quantified according to an established protocol (Lee *et al.*, 2020). B16F10 cells were seeded into 24-well plates (5.0×10^4 cells/well in 500 µL complete medium) and treated with vehicle (D), **1a**, **1b**, **1c**, or **1d** (100 µM for 48 h). For dose-dependency studies, cells were treated with 25, 50, 100, or 200 µM of the respective analogues. After 48 h, the media containing floating cells was aspirated. Adherent cells were washed by gently swirling with 5.0 mL PBS, which was then aspirated. Cells were detached with 0.25 % trypsin (1.0 mL), neutralized with complete media (9.0 mL), and centrifuged at 500 × g for five min. The supernatant was discarded, and the cell pellet was dissolved in 100 µL of 1.0 N NaOH containing 1.0 % DMSO at 80 °C for 1 h. Subsequently, 100 µL of each lysate was transferred to a 96-well plate, and absorbance was measured at 405 nm using a microplate reader. Melanin concentration was determined using a standard curve generated from synthetic melanin (0 to 400 µg/mL) dissolved in 1.0 N NaOH. Phase-contrast images of adherent cells treated with D or **1a** were acquired at 10× magnification using an inverted Nikon microscope.

Lectin staining for imaging by confocal microscopy

B16F10 cells were seeded at a cell density of 5.0×10^4 cells/well in 500 µL complete medium in 24-well plates containing sterile 1.0 mm coverslips (12 mm diameter) and were either left untreated (U) or treated with vehicle (D), **1a**, **1b**, **1c**, or **1d** (100 µM, added to cell suspension before plating). After 48 h, the spent medium was aspirated, and the cells were gently washed with 0.5 mL PBS. Cells were then fixed with 0.5 mL of freshly prepared 4.0 % paraformaldehyde (PFA) in PBS for 10 min at room temperature (RT), followed by three washes with 0.5 mL of 0.1 % PBST (PBS containing 0.1% Tween 20). Cells were permeabilized with 0.5 mL of PBS containing 0.1 % Triton X-100 for 5 min at RT, then washed three times with 0.5 mL of 0.1 % PBST for 2.0 min each. Blocking was performed with 0.5 mL of 2.0 % gelatin in 0.1 %

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PBST for 30 min at RT, followed by six washes with 1.0 mL of 0.1 % PBST (2 min each). Cells were stained with fluorescein-conjugated wheat germ agglutinin (WGA; 1:2000 dilution from a 5.0 mg/mL stock) in 100 µL of 0.1 % PBST for 30 min at RT. Following lectin staining, nuclei were counterstained with DAPI (10 ng/mL in 100 µL PBS for 20 min at RT). Cells were then washed three times with 0.5 mL of 0.1 % PBST for 2.0 min each. One drop of ProLong Gold Anti-fade reagent was placed on a clean glass slide; after draining the residual PBST, coverslips were inverted and gently placed onto the antifade droplet to avoid air bubbles. Slides were cured overnight at 4 °C, and the edges were sealed with transparent nail polish prior to imaging. Samples were imaged on a Zeiss LSM 980 confocal microscope using a Plan-Apochromat 63× oil immersion objective. At least three coverslips (three biological replicates) were analyzed per condition, and three non-overlapping fields were acquired per coverslip. Fluorescence channels were assigned as follows: blue, DAPI; green, fluorescein-WGA.

Lectin blotting for glycan profiling

B16F10 cells were treated with D, **1a**, **1b**, **1c**, or **1d** (100 µM, 48 h) while in suspension, alongside an untreated (U) control. After 48 h, the medium containing non-adherent cells was discarded. Adherent cells were detached with 0.25 % trypsin (1.0 mL), neutralized with complete medium (9.0 mL), and centrifuged at 500 × g for five min. The supernatant was aspirated, and the cell pellet was washed with PBS (3 × 5.0 mL at 3,000 × g for 2.0 min each). Cell lysates were prepared in 100 µL RIPA buffer supplemented with PIC (1:100). Protein concentration was determined using the Bradford assay. Equal amounts of protein were resolved on 7.5 % SDS-PAGE and transferred to nitrocellulose membranes at a constant current of 250 mA for three h at 4 °C. Membranes were blocked for 1 h at RT with 2.0 % gelatin in 0.1 % PBST (10 mL per membrane). Blocked membranes were incubated for 1 h with biotinylated *Maackia amurensis* lectin-II (MAL-II; 1:2000 from a 1.0 mg/mL stock) or fluorescein-conjugated *Vicia villosa* agglutinin (VVA; 1:2000 from a 2.0 mg/mL stock) diluted in 5.0 mL of 0.1 % PBST. Membranes were then washed six times with 10 mL of 0.1 % PBST for five min each. VVA blots were visualized on an Amersham Typhoon imager using the Cy2 channel. For MAL-II detection, membranes were incubated for 30 min with FITC-avidin (1:50,000 from a 2.0 mg/mL stock) in 5.0 mL of 0.1 % PBST, followed by six additional washes with 10 mL of 0.1 % PBST (five min each). MAL-II blots were then imaged on the Amersham Typhoon using the Cy2 filter. Three biological replicates were performed for each lectin.

Lectin flow cytometry

Lectin binding was assessed using a reported protocol (Agarwal *et al.*, 2013). B16F10-Luc2 cells were harvested by standard trypsinization as described above. For treatment, 1.5 mL of complete medium was added to each well of a six-well plate, and wells were left untreated (U) or treated with vehicle (DMSO), **1a**, **1b**, **1c,** or **1d** (100 µM final concentration). Cells (3.0×10^5 cells in 1.5 mL complete medium; final density 1.0×10^5 cells/mL) were added to each well and incubated for 48 h at 37 °C. After treatment, cells were detached with trypsin and collected by centrifugation (500 × g, 5.0 min). Supernatants were discarded, and pellets were resuspended in 5.0 mL complete medium and incubated at 37 °C for 30 min to allow replenishment of cell surface glycans. Cells were then pelleted again (500 × g, 5.0 min) and resuspended

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in 1.0 mL PBS to obtain a density of 2.0 × 10^5 cells/mL. Suspensions were transferred to pre-labelled microcentrifuge tubes (MCTs) and washed twice with 1.0 mL PBS. For neuraminidase controls, 2.0 × 10^5 cells were resuspended in 100 µL PBS and incubated with α2→3,6,8-neuraminidase from *Clostridium perfringens* (final concentration 100 units/mL) at 37 ºC for 30 min, followed by two PBS washes (1.0 mL each). For lectin staining, cells were resuspended in either 500 µL lectin staining buffer (LSB) containing SNA (0.5 µg/mL) or 200 µL LSB containing VVA (10 µg/mL) and incubated for 15 min at 4.0 °C. Cells were washed once with 500 µL LSB, then resuspended in 200 µL LSB, and 0.5 µL propidium iodide (2.5 µg/mL) was added to each sample immediately before acquisition. Flow cytometry data were acquired by sequential gating: FSC-A/SSC-A to exclude debris, FSC-H/FSC-A to exclude doublets, followed by gating on PI-A-negative live cells; 10,000 live events were collected per sample. Two biological replicates per condition, each with two technical replicates, were analyzed using FlowJo v10, and geometric mean fluorescence intensity (GMFI) of FITC-positive populations was plotted.

Immunoprecipitation and Lectin Blotting of Pmel17/gp100

Untreated and **1a**-treated (100 µM, 48 h) B16F10 cells were lysed in RIPA buffer as described previously, and total protein concentration was determined using the Bradford assay. For each immunoprecipitation reaction, 1 mg of total cell lysate was incubated with 1 µg of anti-gp00 (*N*-term) antibody overnight at 4 °C with end-to-end rotation to facilitate antigen-antibody binding. Subsequently, 50 µL of pre-washed Protein G agarose beads (equivalent to 20 µL of bead slurry, Merck; 16-266) were added to the immune complexes. The mixture was incubated for 2 h at 4 °C, followed by an additional 1 h at RT on an end-over-end rotor. Bead-antibody-antigen complexes were pelleted by centrifugation at 1400 rpm for 5 min, and the supernatant was collected as the ‘depleted’ fraction. The beads were washed twice with 250 µL of 0.1 % PBST using gentle wave motion mixing. Immunoprecipitated proteins were eluted twice with 30 µL of 0.2 M Glycine (pH 2.6). Aliquots from all fractions, input, depleted, wash, elution, and boiled beads (BB), were boiled at 95 °C for 10 min. Samples were resolved by SDS-PAGE on a 10 % polyacrylamide gel at 120 V and transferred to a nitrocellulose membrane as described above.

For Pmel17/gp100 detection, the membrane was blocked for one h with 5.0 % (w/v) non-fat milk (NFM) (5.0 mL) in PBS. The blocked membrane was incubated overnight at 4 °C with anti-gp100 (*N*-term) primary antibody (1:10,000 dilution, stock conc. 0.357 mg/mL) in 5.0 mL of 5.0 % (w/v) NFM in PBS. Membrane was washed thrice with 0.1 % PBST (10 mL/wash, 5 min each) and incubated for 1 h at RT with AF488-conjugated anti-rabbit IgG secondary antibody (1:10,000 dilution, stock conc. 2.0 mg/mL) in 5.0 mL 0.1 % PBST. Following three additional washes with 0.1 % PBST (10 mL/wash, 5 min each), signals were visualized using an Amersham Typhoon fluorescent scanner. Glycan analysis was performed using lectin blotting with MAL-II and VVA as described previously. Two biological replicates, each with two technical replicates, were performed for each antibody or lectin probe.

Western Blotting for Melanosomal Proteins

Cell lysates from analogue-treated samples were prepared as described above. Proteins were resolved by 7.5 % SDS-PAGE and transferred to a nitrocellulose

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membrane using a wet transfer system at a constant current of 250 mA for three h at 4 °C. Membranes were blocked for one h at RT with 5.0 % (w/v) NFM (5.0 mL) in PBS. Following blocking, membranes were incubated overnight at 4 °C with the following primary antibodies diluted in 5.0 mL of 5 % (w/v) NFM in PBS: anti-gp100 (*N*-term; 1:10,000, stock conc. 0.357 mg/mL); anti-gp100 (HMB45; 1:1000, stock conc. 0.2 mg/mL); anti-gp100 (C-term; 1:1000, stock conc. 0.5 mg/mL); anti-tyrosinase (1:1000, stock conc. 0.2 mg/mL); anti-MART-1 (1:5000, stock conc 1.08 mg/mL); and anti-β-actin (1:20,000, stock conc. 2.0-2.5 mg/mL). Membranes were washed thrice with 0.1 % PBST (10 mL) for five min each. Subsequently, membranes were incubated for 1 h at RT with the corresponding 2° antibodies (diluted 1:10,000 in 5.0 mL of 0.1 % PBST). AF488-conjugated anti-rabbit IgG (stock conc. 2.0 mg/mL) was used for gp100 (*N*-term), tyrosinase, and MART-1 detection. Cy5-conjugated donkey anti-mouse IgG (stock conc. 1.4 mg/mL) was used for gp100 (HMB45) and β-actin, while AF488-conjugated anti-mouse IgG (stock conc. 2.0 mg/mL) was used for gp100 (*C*-term). After three final washes with 0.1 % PBST (10 mL, five min each), fluorescent signals were visualized using an Amersham Typhoon scanner. Experiments were conducted with two biological replicates and two technical replicates per antibody probe.

For the dose dependency study of gp100 (*N*-term) and gp100 (HMB45), cells were either left untreated (U) or treated with vehicle (D), **1a**, **1b**, **1c**, or **1d** at concentrations of 25, 50, 100, and 200 µM. For the time-course study, B16F10 cells were either left untreated (U) or treated with vehicle (D), **1a**, **1b**, **1c**, or **1d** (100 µM) for various time points (24, 48, 72, and 96 h). The remainder of the protocol was performed as described above. Two biological replicates, each with two technical replicates, were performed for this experiment.

Staining of Pmel17/gp100 for Confocal Microscopy

B16F10 cells were analogue-treated, fixed, and permeabilized as described previously. The cells were washed three times with 0.5 mL of 0.1 % PBST for 2.0 min each. For F-actin staining, cells were incubated with 66 nM rhodamine phalloidin (0.2 µL; stock: 66 µM in DMSO) in 200 µL of 3.0 % BSA in PBS for 20 min at RT. Following staining, cells were washed thrice with 0.5 mL of 0.1 % PBST for two min each. Cells were incubated with primary antibodies in 0.1 mL of PBS overnight at 4 °C. The following primary antibodies were used: unconjugated anti-gp100 (*N*-term) (dilution 1:100, stock conc. 0.357 mg/mL) or anti-gp100 (HMB45) (dilution 1:100, stock conc. 0.2 mg/mL). The cells were washed thrice with 0.5 mL of 0.1 % PBST for five min each. Subsequently, cells were incubated with the corresponding secondary antibodies in 0.1 mL of PBS for 2 h at RT in the dark: Alexa Fluor 488 anti-rabbit IgG (dilution 1:1,000, stock conc. 2.0 mg/mL) for gp100 (*N*-term) or Cy5-conjugated donkey anti-mouse IgG (dilution 1:1,000, stock conc. 1.4 mg/mL) for gp100 (HMB45). After staining, cells were washed three times with 0.5 mL of 0.1 % PBST for 2.0 min each. Following antibody staining, cells were incubated with DAPI (10 ng/mL) in 100 µL of PBS for 20 min at RT. Cells were washed three times with 0.5 mL of 0.1 % PBST for 2.0 min each. Slides were mounted and visualized as described previously. At least three coverslips, representing three biological replicates, were scanned. For each biological replicate, three fields were imaged per coverslip. Fluorescence channels were assigned as follows: DAPI (Blue), Rhodamine Phalloidin (Red), Alexa Fluor 488 for Pmel17/gp100 (*N*-term) (Green), and Cy5 for Pmel17/gp100 (HMB45) (Red).

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Colocalization of Melanosomal Proteins by Confocal Microscopy

B16F10 cells were **1a**-treated, fixed, and permeabilized as described previously. Cells were incubated with unconjugated anti-MART-1 (clone 48-118) antibody (dilution 1:100, stock conc. 1.04 mg/mL) in 0.1 mL of PBS overnight at 4 °C. The cells were washed thrice with 0.5 mL of 0.1 % PBST for five min each. Subsequently, cells were stained with Alexa Fluor 488 anti-rabbit IgG (dilution 1:1,000, stock conc. 2.0 mg/mL) in 0.1 mL of PBS for 2 h at RT in the dark. The cells were then washed thrice with 0.5 mL of 0.1 % PBST for two min each. Following MART-1 staining, cells were incubated with conjugated Alexa Fluor 647 anti-gp100 (*N*-term) antibody (dilution 1:100, stock 0.5 mg/mL) in 0.1 mL of PBS overnight at 4 °C. The cells were washed thrice with 0.5 mL of 0.1 % PBST for five min each. For nuclear counterstaining, DAPI (10 ng/mL) was added in 100 µL of PBS and incubated for 20 min at RT. Cells were washed three times with 0.5 mL of 0.1 % PBST for two min each. Slides were mounted and visualized as described previously. At least three coverslips, representing three biological replicates, were scanned. For each biological replicate, three fields were imaged per coverslip. DAPI (Blue), Alexa Fluor 488 for MART-1 (Green), and Alexa Fluor 647 for Pmel17/gp100 (*N*-term) (Red). Yellow signals indicate the colocalization of MART-1 with Pmel17/gp100.

Quantification of Siglec-E-Fc Binding by Flow Cytometry

B16F10-Luc2 cells were harvested using standard trypsinization protocols as previously described. Cells (1.0×10^6) were seeded in 5.0 mL of complete media (final cell density of 1.0×10^5 per mL) into 10-cm tissue-culture treated plates. Plates were either left untreated or treated with vehicle (DMSO, D), **1a** (100 µM), **1b** (100 µM), **2a** (100 µM), or Dox (100 nM) and incubated for 48 h. Following incubation, cells were trypsinized, harvested, and washed twice with PBS (10 mL). Aliquots of 1.0×10^6 cells were prepared for analysis. For the negative control, 1.0×10^6 cells were resuspended in 100 µL of PBS in an MCT and incubated with α2→3,6,8-neuraminidase (final concentration of 100 units/mL) at 37 ºC for 10 min. Cells were subsequently washed twice with PBS (1.0 mL). A 50 µL antibody pre-complex was prepared by mixing recombinant Siglec-E-Fc chimera (final concentration of 75 nM), Alexa Fluor 488-conjugated goat anti-mouse IgG secondary antibody (final concentration of 50 nM), and 1.0 µL of AmCyan live/dead stain in PBS in an MCT. The pre-complex was incubated at 4 °C for 30 min. This pre-complex was then added to the cell aliquots and incubated for 30 min at 4 °C. A secondary antibody-only control was included to account for non-specific binding. Cells were washed twice with FACS buffer (1.0 mL), resuspended in 300 µL of FACS buffer, and analyzed. Cells were gated sequentially on FSC-A/SSC-A (to exclude debris), FSC-H/FSC-A (for doublet exclusion), and AmCyan-A (to select live cells based on negative staining). For each sample, 10,000 live events were acquired. Three biological replicates were prepared, with two technical replicates (two counts per sample) for each BR. Siglec-E-Fc binding was analyzed using FlowJo software, and the MFI was plotted for the AF488-positive population.

Migration and Invasion Assays

Migration assays were performed following a reported protocol (Partsch and Schwarzer, 1991). B16F10-Luc2 cells were harvested using standard trypsinization

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protocols as previously described. The resulting cell pellet was washed twice with PBS (10 mL). 1.0 × 10^5 cells were resuspended in complete media (0.3 mL) and seeded into trans-well inserts equipped with an 8.0 µm pore size PET membrane for migration. For invasion, the trans-well inserts were pre-coated with Matrigel, mimicking the extracellular matrix. The inserts were carefully placed into a 24-well plate using forceps, establishing a two-chamber system: the upper chamber (insert) containing the cells (0.3 mL), and the bottom chamber (well) filled with complete media (0.7 mL) as a chemoattractant. Cells in the upper chamber were either left untreated (U) or treated with vehicle (D), **1a** (100 µM), or **1b** (100 µM) in a total volume of 1.0 mL. After cells adhered to the insert membrane (~ 12 h), the complete media in the upper chamber was replaced with incomplete media (without FBS). Simultaneously, the media in the bottom chamber was replaced with fresh complete media to establish a chemotactic gradient. Cells were then either left untreated (U) or replenished with vehicle (D), **1a** (100 µM), or **1b** (100 µM) in a total volume of 1.0 mL and incubated for 48 h. Media was aspirated from the upper and bottom chambers, and the inserts were washed once by dipping them in a 100 mL beaker filled with PBS. The inserts were then placed into fresh wells, and 0.5 mL of 0.25 % aq. crystal violet stain was added to both the upper and bottom chambers. After a 20 min incubation, the inserts were washed twice by dipping them into a PBS-filled 100 mL beaker. Non-migrated cells on the interior of the upper chamber were removed using a cotton-tipped pipette without damaging the membrane. The upper chambers were placed directly on an inverted bright-field microscope, and images were acquired using a 10× objective. Subsequently, the membranes were excised from the inserts using a surgical blade and placed into a 96-well plate. The membranes were soaked in 100 µL of DMSO to solubilize the stained cells, and absorbance was measured at 570 nm. Three biological replicates were prepared, with two technical replicates for each BR.

For the dose dependency study, inserts were either left untreated (U) or treated with vehicle (D), **1a**, or **1b** at 50, 100, and 200 µM concentrations. The remainder of the protocol was performed as described above. Two biological replicates, each with two technical replicates, were prepared for this experiment.

Metabolic glycan engineering (MGE) to test tumor accessibility of GalNAc analogues

Optimization of MGE conditions in melanoma cells was performed following a reported protocol prior to *in vivo* testing (Hang *et al.*, 2003). B16F10-Luc2 cells were analogue-treated for 48 h and harvested using standard trypsinization as previously described. At the indicated time point, cells were counted, and aliquots of 1.0 × 10^6 cells were pelleted by centrifugation (500 × g, 5.0 min). The resulting pellets were washed with PBS (10 mL). Cell pellets were resuspended in 200 µL of PBS and subjected to a SPAAC reaction with 10 µM DBCO-Cy5 for 30 min. Two technical replicates were performed for each condition. Following the reaction, cells were immediately lysed by adding 500 µL of a chloroform:methanol:water (1:4:3) mixture. Proteins remaining in the solution were precipitated using methanol to yield a protein pellet. This pellet was resuspended in methanol and washed three times (centrifugation at 16000 × g, 5 min). Finally, the protein pellets were resuspended in 100 µL of 1.0 % SDS buffer. Protein concentration was determined using the Bradford assay. Approximately 20 µg of protein was loaded onto 7.5 % SDS-PAGE gels and resolved until the tracking dye reached the bottom of the gel. The gels were washed

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once with PBS (10 mL) in a glass container. Subsequently, the gels were placed in an Amersham Typhoon fluorescence imager and directly visualized using the Cy5 (red) channel. Silver-stained gels, run in parallel, served as loading controls. Two biological replicates, each with two technical replicates, were performed for each condition.

In vivo metabolic engineering using **1e** was assessed following a previously described protocol (Laughlin *et al.*, 2006). Mice received intraperitoneal injections of either 300 mg/kg of **1e** or vehicle (D) once on days one, three, five, seven, and nine. On day 10, mice were euthanized, and tumors were excised and homogenized. Cell suspensions were prepared and processed for protein extraction and SPAAC labelling as described above. Three biological replicates, derived from three distinct mouse samples (numbered 1, 2, and 3), were analyzed with two technical replicates for each.

Implantation of Melanoma Cells Followed by Intraperitoneal HexNAc Analogue Administration

B16F10-Luc2 cells were harvested using standard trypsinization protocols as previously described. Cells were counted, and aliquots of 1.0×10^6 cells were pelleted by centrifugation (500 × g, 5 min). Pellets were washed twice with PBS (10 mL). For implantation, cells (1.0×10^6) were resuspended in 50 µL of phosphate-buffered saline (PBS) and injected subcutaneously using a 27 ½-inch gauge syringe needle.

DMSO vs **1a**

Following subcutaneous tumor cell implantation, mice (n = 9; three per group in three independent experiments) received intraperitoneal injections of either 100 µL of vehicle DMSO (D) or **1a** (200 mg/kg). Injections were administered once on days one, three, five, seven, and nine post-implantation. Analogues were administered under anaesthesia whenever the dosing day coincided with an imaging session.

1a vs **1b**

Following subcutaneous tumor cell implantation, mice (five per group) received intraperitoneal injections of either 100 µL of **1a** (200 mg/kg) or **1b** (200 mg/kg). Injections were administered once on days one, three, five, seven, and nine post-implantations. Analogues were administered under anesthesia whenever the dosing day coincided with an imaging session.

1a vs Dox

Following subcutaneous tumor cell implantation, mice (five per group) received intraperitoneal injections of either 100 µL of **1a** (200 mg/kg) or Dox (2.0 mg/kg). **1a** was administered once, on days one, three, five, seven, and nine under anesthesia. **Dox** was administered once on days one, seven, and 14. Analogues were administered under anesthesia whenever the dosing day coincided with an imaging session.

1b vs **2b**

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Following subcutaneous tumor cell implantation, mice (four per group) received intraperitoneal injections of either 100 µL of **1b** (200 mg/kg) or **2b** (200 mg/kg). Injections were administered once on days one, three, five, seven, and nine post-implantation. Analogues were administered under anesthesia whenever the dosing day coincided with an imaging session.

Tumor Size and Volume Measurement

Tumor growth was monitored in real-time using BLI following treatment with the substrate D-luciferin (stock concentration of 15 mg/mL in PBS). Mice were anesthetized and injected i.p. with 150 µL of D-luciferin (1.5 mg per mouse). Imaging was performed using an *in vivo* imaging system (Lago X). To prevent signal saturation, the exposure time was fixed at 60 s, and binning was adjusted from low to medium. Bioluminescence intensity was quantified in photons/sec using Aura software (version 2.03). Tumor dimensions were measured at regular intervals using digital vernier calipers. Tumor volume was calculated using the formula [tumor volume in mm^3 = (length × $width^2$)/2]. Body weight was measured and recorded every alternate day, beginning on day one and continuing until the end of the study.

Intravenous Administration of B16F10-Luc2 Cells

B16F10-Luc2 cells were harvested using standard trypsinization protocols as previously described. Cells (1.0×10^6) were seeded in 5.0 mL of complete media (final cell density of 1.0×10^5 per mL) into 10-cm tissue-culture treated plates containing 5.0 mL of complete media. Plates were either left untreated (U) or treated with vehicle (D), **1a** (100 µM), or **1b** (100 µM) and incubated for 48 h. Following incubation, cells were detached by trypsinization and washed twice with PBS (10 mL). Cells were counted, and aliquots of 5.0×10^5 cells were pelleted by centrifugation (500 × g, 5.0 min). The final cell pellet was resuspended in 50 µL of PBS. The cell suspension was administered via the retro-orbital plexus using a 27½ inch-gauge syringe needle. Injections were performed at an acute angle with the bevel facing up at a slow rate to minimize injury. Four mice were included per group for this study.

Harvesting, Bioluminescence Imaging, and Fixation of Lungs

Mice were euthanized via CO_2 asphyxiation on day 19. The lungs and heart were dissected *en bloc* and immersed in sterile PBS (10 mL) to remove residual blood. For imaging, the lungs were placed in a 3.5-cm diameter dish and incubated with 100 µL of D-luciferin (15 mg/mL). BLI was acquired with an exposure time of 20 s and a medium binning setting. Following imaging, the lungs were rinsed in sterile PBS to remove residual D-luciferin. The tissues were then stored in glass vials containing 10 mL of Fekete's solution (Dong, Maziveyi and Alahari, 2015) (prepared using 100 mL of 70 % alcohol, 10 mL of 10 % formalin, and 5.0 mL of glacial acetic acid). Under these conditions, melanoma nodules appeared black against the pale white background of the lung tissue. Nodules were also observed in the cervical lymph nodes. High-definition images were captured using a 64-megapixel smartphone camera.

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MTT assay for cell viability

This assay measures cellular metabolic activity as an indicator of cytotoxicity (Mosmann, 1983). It is a colorimetric assay based on the reduction of the yellow tetrazolium salt (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT), to purple formazan crystals by metabolically active cells. Viable cells contain NAD(P)H-dependent oxidoreductase enzymes that catalyze this reduction. The insoluble formazan crystals are dissolved, and the resulting-colored solution was quantified by measuring absorbance. B16F10 cells were seeded in 96-well flat bottom plates (5.0×10^4 cells per mL) in 200 µL of complete medium with phenol red (10,000 cells/well). Cells were either left untreated (U), or treated with vehicle (D), **1a, 1b, 1c,** or **1d** while in suspension at concentrations of 25, 50, 100, and 200 µM for 48 h at 37 °C. After 48 h, 10 µL/well of MTT labelling reagent (stock solution: 5.0 mg/mL in water; final concentration 0.5 mg/mL) was added to each well. The cells were incubated with MTT for 4 h at 37 °C, after which purple formazan crystals were visually confirmed under the microscope. The media was aspirated from all wells, and 100 µL/well of DMSO was added to dissolve the formazan crystals. The crystals were resuspended thoroughly using a multi-channel pipette, and absorbance was measured at 570 nm using a microplate reader.

Trypan Blue Staining Assay

Trypan blue staining is a widely used method for measuring cell viability (Strober, 2015). Trypan blue is an azo dye impermeable to intact cell membranes; therefore, it enters only cells with compromised membranes (dead cells). Upon entry, the dye binds to intracellular proteins, staining the cells a distinctive blue colour. This allows for the differentiation of live (unstained) and dead (blue) cells within a population. B16F10 cells were seeded in a 24-well plate (5.0×10^4 cells per mL/well) in 500 µL of complete medium (25k cells/well). Cells were either left untreated (U) or treated with vehicle (D), **1a, 1b, 1c,** or **1d** (100 µM, 48 h) while in suspension. After 48 h, the media containing floating cells was collected into fresh MCT. Adherent cells were detached by incubation with 0.25 % trypsin (500 µL), followed by dilution with complete media (2.0 mL). The floating cell fraction and the trypsinized cell suspension were combined and centrifuged at 500 × g for five min. The resulting cell pellet was resuspended in 1.0 mL of PBS. For counting, 5.0 µL of 0.4 % (w/v) trypan blue (40 mg trypan blue powder in 10 mL PBS) was mixed with 5.0 µL of the cell suspension and incubated for three min at RT. A 10 µL aliquot of the mixture was loaded onto a hemocytometer. Cells were counted in the four large corner squares and the central square under a microscope. Viable (unstained) and non-viable (stained) cells were counted separately. The total cell count across five squares was approximately 120 ± 15 cells. The total number of viable cells per mL was calculated by multiplying the total number of viable cells counted by the dilution factor. The total cell count per mL was determined by summing the viable and non-viable cells from the five squares. Cell viability was calculated using the following formula: Viable cells (%) = (total number of viable cells/total number of cells) × 100.

Assay for cell proliferation

The stable incorporation of the intracellular fluorescent dye 5-(and -6)-carboxyfluorescein diacetate succinimidyl ester (CFSE) monitors cell division, as

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fluorescent labelling decreases sequentially in daughter cells (Parish *et al.*, 2009). B16F10 cells at 0 h were resuspended in PBS containing 5.0 % FBS at a concentration of 5.0 × 10^4 cells/mL. To label the cells, 1.0 µL of CFSE solution (stock 5.0 mM; 2.8 mg CFSE in 1.0 mL DMSO) was added to 1.0 mL of the cell suspension to achieve a final concentration of 5.0 µM, followed by rapid mixing. After a five min incubation at RT, 10 mL of PBS containing 5.0% FBS was added to quench the reaction. Cells were centrifuged for five min at 300 × g at 20 °C, and the supernatant was aspirated. The cells were washed thrice by resuspending in 1.0 mL of PBS containing 5.0 % FBS. Following staining, B16F10 cells were seeded in 100 mm diameter tissue-culture treated dishes (5.0 × 10^4 cells/mL) in 10 mL of complete medium. Cells were either left untreated (U) or treated with vehicle (D), **1a, 1b, 1c,** or **1d** (100 µM, 48 h) while in suspension. After 48 h, media containing any floating cells was aspirated. Adherent cells were washed by adding PBS (5.0 mL), swirling gently, and aspirating. Cells were detached by incubating with 0.25 % trypsin (1.0 mL), followed by dilution with complete media (9.0 mL). The cell suspension was centrifuged at 500 × g for five min, and the supernatant was discarded. The same staining procedure described for 0 h was performed for the treated cells. Data were acquired using a BD FACSVerse flow cytometer and analyzed using FlowJo V10 software. Two biological replicates were prepared for each condition, and each sample was counted twice, with 10,000 events acquired per run.

RNA isolation and real-time qPCR analysis

B16F10 cells were seeded at a density of 5.0 × 10^4 cells/mL in 10 mL of complete medium in 100 mm diameter tissue-culture treated dishes. Cells were either left untreated (U) or treated with vehicle (D), **1a**, **1b**, **1c**, or **1d** (100 µM) for 48 h. Following incubation, the media was aspirated. Cells were detached by incubating with 0.25 % trypsin (1.0 mL) and diluted with complete media (9.0 mL). The suspension was centrifuged at 500 × g for 5 min at RT, and supernatants were discarded. RNA was isolated using the TRIzol method(Rio *et al.*, 2010). The cell pellets were resuspended in 750 µL of TRIzol reagent and incubated for five min to ensure complete dissociation of nucleoprotein complexes. Subsequently, 150 µL of chloroform was added, incubated for three min, and centrifuged at 12000 × g for 15 min at 4 °C. The colorless upper aqueous phase containing the RNA was transferred to a fresh tube. To precipitate RNA, 375 µL of isopropanol was added and mixed by pipetting. After a 10 min incubation at RT, samples were centrifuged at 12000 × g for 10 min at 4 °C, yielding a white gel-like pellet. The supernatant was discarded, and the pellet was washed by resuspension in 750 µL of 75 % aq. ethanol. Samples were vortexed briefly and centrifuged at 7,500 × g for five min at 4 °C. The supernatant was discarded, and the pellet was air-dried for 10 min in a fume hood. Finally, the RNA was resuspended in 50 µL of RNAse-free water, incubated at 55 °C for 15 min, and stored at -80 °C. RNA concentration and purity (260/280 ratio) were measured using a Biotek reader (1 µL sample volume). RNA integrity was verified by 1.0 % agarose gel electrophoresis. Subsequently, 1.0 µg of RNA was reverse transcribed into cDNA using a cDNA synthesis kit (Invitrogen, 18080051). Quantitative real-time (RT-qPCR) was performed on a QuantStudio 6 Flex system using SYBR fast and gene-specific primers (**Table 1**) to amplify melanosomal and housekeeping genes. The cycle number crossing the threshold was defined as the threshold cycle (Ct) (Gerard *et al.*, 1998). Gene expression was normalized by calculating ΔCt (Ct of target melanosomal genes – Ct of housekeeping gene). Relative quantification between GalNAc analogue-treated

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cells and vehicle-treated controls was performed using the comparative Ct method ($2^{-\Delta\Delta Ct}$) (Schmittgen and Livak, 2008). The fold change relative to the vehicle control (D) was plotted.

Table 1. List of nucleotide sequences of primers for melanosomal-specific genes used in RT-qPCR analysis

Gene Name	Sequence (5' to 3')
FP_gp100 full length (mouse)	TACTGGATGACACCGACACC
RP_gp100 full length(mouse)	ATCCCCTTCACTGAATGGCA
FP_gp100 M-alpha (mouse)	GGCACACACACAATGGAAGT
RP_gp100 M-alpha (mouse)	AGGAAGTGCTTGGTCTCTCC
FP_gp100 M-beta (mouse)	TGAAGGGGATGCATTTGAGC
RP_gp100 M-beta (mouse)	GCAATATGTCCCTGAGCCAC
FP_Tyrosinase (mouse)	CTCTGGGCTTAGCAGTAGGC
RP_Tyrosinase (mouse)	GCAAGCTGTGGTAGTCGTCT
FP_MITF (mouse)	CCAACAGCCCTATGGCTATGC
RP_MITF (mouse)	CTGGGCACTCACTCTCTGC
FP_MART-1 (mouse)	CTGCTGGTACTGTAGAAGACGA
RP_MART-1 (mouse)	GGCTCTCACATGAGCATCTTTC
FP_β-Actin (mouse)	GGCTGTATTCCCCTCCATCG
RP_β-Actin (mouse)	CCAGTTGGTAACAATGCCATGT

RNA sequencing, library preparation, and data analysis

B16F10 cells were treated with D or **1a** (100 μM, 48 h) as described above. Following treatment, total RNA was extracted using the TRIzol method as previously described. RNA concentration and quality were assessed using a spectrophotometer. The isolated RNA was then subjected to high-throughput sequencing to obtain expression data.

The concentration of isolated RNA was quantified using a Qubit 4 Fluorometer with the Qubit HS RNA Assay kit, suitable for concentrations ranging from 0.2 to 200 ng/μL. RNA integrity was assessed using and Agilent 5200 Fragment Analyzer, and an RNA Quality Number (RQN) score was assigned to each sample. Only samples with an RQN value > 8 were selected for library preparation. RNA sequencing libraries were constructed using the QIAseq FastSelect RNA Library Kit, which features cDNA barcoding and rRNA removal, starting with 1 ng to 1000 ng of total RNA. Following construction, the quality of the final libraries was verified on the Fragment Analyzer. The libraries were then pooled and sequenced on an Illumina NextSeq 2000 platform.

Raw sequencing data was processed using fastp to perform quality control and trimming, generating high-quality trimmed data. The processed reads were then aligned to the reference Ensembl mouse genome using HISAT2. Gene expression levels were quantified by assigning mapped reads to genomic features using featureCounts. Differential expression (DE) analysis was subsequently performed using DESeq2 package to identify genes with significant expression changes between experimental conditions. Downstream analysis, including pathway enrichment and visualization, was conducted using various R packages. DEGs were identified and filtered out with the following criteria: false discover rate < 0.05 and |log2Fc| > 1.5.

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PCR Array analysis for genes involved in cancer metastasis

B16F10 cells were seeded at a density of 5.0 × 10^4 cells/mL in 10 mL of complete medium in 100 mm diameter tissue-culture treated dishes. Cells were treated with either vehicle (D) or **1a** (100 μM) for 48 h while in suspension. Following incubation, RNA was isolated and converted to cDNA (starting with approximately 1 μg of RNA) as described previously. The prepared cDNA (50 ng per sample) was analyzed using mouse tumor metastasis RT^2 Profiler PCR array plates. These plates contained pre-coated primers for 84 selected genes involved in metastasis, including those regulating cell adhesion, extracellular matrix components, cell cycle, cell growth and proliferation, apoptosis, and transcription factors (**Table 2**). For the assay, 50 ng/μL cDNA was added to each well, and quantitative real-time PCR was performed using the KAPA SYBR fast qPCR kit (KAPA Biosystems, KK4602). Differences between **1a**-treated cells and vehicle-treated controls were quantified using the comparative Ct method (2^{-}ΔΔCt), as mentioned previously. Results were visualized as a Volcano plot, displaying the relationship between fold change and statistical significance. The x-axis represents fold change calculated as log2(2^{-}ΔΔCt), and the y-axis represents statistical significance calculated as -log10(P-value)(Colitti and Stefanon, 2016).

Table 2. List of housekeeping (control) and metastasis-related genes included in the selected PCR array, with descriptions and GenBank IDs

Position	UniGene	GenBank	Symbol	Description
A01	Mm.384171	NM_007462	Apc	Adenomatosis polyposis coli
A02	Mm.29628	NM_134155	Brms1	Breast cancer metastasis-suppressor 1
A03	Mm.341574	NM_013654	Ccl7	Chemokine (C-C motif) ligand 7
A04	Mm.423621	NM_009851	Cd44	CD44 antigen
A05	Mm.4261	NM_007656	Cd82	CD82 antigen
A06	Mm.35605	NM_009864	Cdh1	Cadherin 1
A07	Mm.1571	NM_009866	Cdh11	Cadherin 11
A08	Mm.57048	NM_007666	Cdh6	Cadherin 6
A09	Mm.441131	NM_007667	Cdh8	Cadherin 8
A10	Mm.4733	NM_009877	Cdkn2a	Cyclin-dependent kinase inhibitor 2A
A11	Mm.333388	NM_145979	Chd4	Chromodomain helicase DNA binding protein 4
A12	Mm.181021	NM_009932	Col4a2	Collagen, type IV, alpha 2
B01	Mm.795	NM_007778	Csf1	Colony stimulating factor 1 (macrophage)
B02	Mm.7286	NM_013502	Ctbp1	C-terminal binding protein 1
B03	Mm.18962	NM_009818	Ctnna1	Catenin (cadherin associated protein), alpha 1
B04	Mm.272085	NM_007802	Ctsk	Cathepsin K
B05	Mm.930	NM_009984	Ctsl	Cathepsin L
B06	Mm.303231	NM_021704	Cxcl12	Chemokine (C-X-C motif) ligand 12
B07	Mm.234466	NM_009909	Cxcr2	Chemokine (C-X-C motif) receptor 2
B08	Mm.1401	NM_009911	Cxcr4	Chemokine (C-X-C motif) receptor 4
B09	Mm.28549	NM_026603	Denr	Density-regulated protein
B10	Mm.262194	NM_015779	Elane	Elastase, neutrophil expressed
B11	Mm.250981	NM_010142	Ephb2	Eph receptor B2
B12	Mm.5025	NM_008815	Etv4	Ets variant gene 4 (E1A enhancer binding protein, E1AF)
C01	Mm.142822	NM_007968	Ewsr1	Ewing sarcoma breakpoint region 1

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C02	Mm.27365	NM_001081286	Fat1	FAT tumor suppressor homolog 1 (Drosophila)
C03	Mm.276715	NM_008011	Fgfr4	Fibroblast growth factor receptor 4
C04	Mm.3291	NM_008029	Flt4	FMS-like tyrosine kinase 4
C05	Mm.193099	NM_010233	Fn1	Fibronectin 1
C06	Mm.1870	NM_008761	Fxyd5	FXYD domain-containing ion transport regulator 5
C07	Mm.302602	NM_053110	Gpnmb	Glycoprotein (transmembrane) nmb
C08	Mm.267078	NM_010427	Hgf	Hepatocyte growth factor
C09	Mm.265786	NM_152803	Hpse	Heparanase
C10	Mm.334313	NM_008284	Hras1	Harvey rat sarcoma virus oncogene 1
C11	Mm.20801	NM_016865	Htatip2	HIV-1 tat interactive protein 2, homolog (human)
C12	Mm.268521	NM_010512	Igf1	Insulin-like growth factor 1
D01	Mm.1410	NM_008360	Il18	Interleukin 18
D02	Mm.222830	NM_008361	Il1b	Interleukin 1 beta
D03	Mm.179747	NM_008398	Itga7	Integrin alpha 7
D04	Mm.87150	NM_016780	Itgb3	Integrin beta 3
D05	Mm.458301	NM_178260	Kiss1	KiSS-1 metastasis-suppressor
D06	Mm.191035	NM_053244	Kiss1r	KISS1 receptor
D07	Mm.383182	NM_021284	Kras	V-Ki-ras2 Kirsten rat sarcoma viral oncogene homolog
D08	Mm.390681	NM_175116	Lpar6	Lysophosphatidic acid receptor 6
D09	Mm.275003	NM_023061	Mcam	Melanoma cell adhesion molecule
D10	Mm.22670	NM_010786	Mdm2	Transformed mouse 3T3 cell double minute 2
D11	Mm.86844	NM_008591	Met	Met proto-oncogene
D12	Mm.14126	NM_019471	Mmp10	Matrix metallopeptidase 10
E01	Mm.4561	NM_008606	Mmp11	Matrix metallopeptidase 11
E02	Mm.5022	NM_008607	Mmp13	Matrix metallopeptidase 13
E03	Mm.29564	NM_008610	Mmp2	Matrix metallopeptidase 2
E04	Mm.4993	NM_010809	Mmp3	Matrix metallopeptidase 3
E05	Mm.4825	NM_010810	Mmp7	Matrix metallopeptidase 7
E06	Mm.4406	NM_013599	Mmp9	Matrix metallopeptidase 9
E07	Mm.212577	NM_054081	Mta1	Metastasis associated 1
E08	Mm.215481	NM_144800	Mtss1	Metastasis suppressor 1
E09	Mm.2444	NM_010849	Myc	Myelocytomatosis oncogene
E010	Mm.1055	NM_008506	Mycl1	V-myc myelocytomatosis viral oncogene homolog 1, lung carcinoma derived (avian)
E011	Mm.297109	NM_010898	Nf2	Neurofibromatosis 2
E012	Mm.439702	NM_008704	Nme1	Non-metastatic cells 1, protein (NM23A)
F01	Mm.1260	NM_008705	Nme2	Non-metastatic cells 2, protein (NM23B)
F02	Mm.41787	NM_019731	Nme4	Non-metastatic cells 4, protein
F03	Mm.247261	NM_015743	Nr4a3	Nuclear receptor subfamily 4, group A, member 3
F04	Mm.1359	NM_011113	Plaur	Plasminogen activator, urokinase receptor
F05	Mm.22347	NM_008891	Pnn	Pinin
F06	Mm.245395	NM_008960	Pten	Phosphatase and tensin homolog
F07	Mm.273862	NM_009029	Rb1	Retinoblastoma 1
F08	Mm.485649	NM_146095	Rorb	RAR-related orphan receptor beta
F09	Mm.311972	NM_011029	Rpsa	Ribosomal protein SA
F10	Mm.335942	NM_023871	Set	SET nuclear oncogene
F11	Mm.391091	NM_010754	Smad2	MAD homolog 2 (Drosophila)
F12	Mm.100399	NM_008540	Smad4	MAD homolog 4 (Drosophila)
G01	Mm.22845	NM_009271	Src	Rous sarcoma oncogene
G02	Mm.454968	NM_009217	Sstr2	Somatostatin receptor 2

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G03	Mm.375031	NM_011518	Sykb	Spleen tyrosine kinase
G04	Mm.252156	NM_013836	Tcf20	Transcription factor 20
G05	Mm.248380	NM_011577	Tgfb1	Transforming growth factor, beta 1
G06	Mm.206505	NM_011594	Timp2	Tissue inhibitor of metalloproteinase 2
G07	Mm.4871	NM_011595	Timp3	Tissue inhibitor of metalloproteinase 3
G08	Mm.255607	NM_080639	Timp4	Tissue inhibitor of metalloproteinase 4
G09	Mm.1062	NM_009425	Tnfsf10	Tumor necrosis factor (ligand) superfamily, member 10
G10	Mm.222	NM_011640	Trp53	Transformation related protein 53
G11	Mm.173847	NM_011648	Tshr	Thyroid stimulating hormone receptor
G12	Mm.282184	NM_009505	Vegfa	Vascular endothelial growth factor A
H01	Mm.328431	NM_007393	Actb	Actin, beta
H02	Mm.163	NM_009735	B2m	Beta-2 microglobulin
H03	Mm.343110	NM_008084	Gapdh	Glyceraldehyde-3-phosphate dehydrogenase
H04	Mm.3317	NM_010368	Gusb	Glucuronidase, beta
H05	Mm.2180	NM_008302	Hsp90ab1	Heat shock protein 90 alpha (cytosolic), class B member 1
H06	N/A	SA_00106	MGDC	Mouse Genomic DNA Contamination
H07	N/A	SA_00104	RTC	Reverse Transcription Control
H08	N/A	SA_00104	RTC	Reverse Transcription Control
H09	N/A	SA_00104	RTC	Reverse Transcription Control
H10	N/A	SA_00103	PPC	Positive PCR Control
H11	N/A	SA_00103	PPC	Positive PCR Control
H12	N/A	SA_00103	PPC	Positive PCR Control

Statistical Analysis

Statistical analyses were performed using GraphPad Prism version 8.4.3 (GraphPad Software, San Diego, USA). Data are presented as mean ± standard deviation (SD). For datasets with a Gaussian distribution, comparisons between two groups were analyzed using multiple t-tests corrected with the Holm-Sidak method. Comparisons among multiple groups were analyzed using one-way or two-way ANOVA followed by Tukey's multiple comparisons test. For datasets with non-gaussian distribution, comparisons between multiple groups were analyzed using Kruskal-Wallis test with Dunn's multiple comparisons test. Survival analysis was determined using the log-rank (Mantel-Cox) test. A p-value of less than 0.05 was considered statistically significant in most cases. Levels of significance are indicated in the figure legends where applicable.

Reagents, Antibodies, and Chemicals

List of reagents used in this study:

List of reagents	Source	Identifier (Cat. /Ref.)
Cell lines		
B16F10	Dr. R. Gokhale, NII	--
B16F10-Luc2	Dr. S. Singha, NII	--
Peracetyl HexNAc Analogues		
Ac_5GalNTGc (1a)	In-house	
Ac_4GalNAc (1b)	-do-	

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Ac_5GalNGc (1c)	-do-	
Ac_3BnGalNAc (1d)	-do-	
Ac_4GalNAz (1e)	-do-	
Ac_5GlcNTGc (2a)	-do-	
Ac_4GlcNAc (2b)	-do-	
Antibodies		
Rabbit monoclonal anti-melanoma gp100 antibody (*N*-term)	Abcam	ab137078
Mouse monoclonal anti-melanoma gp100 antibody (HMB45)	-do-	ab787
Rabbit monoclonal anti-melanoma gp100 Alexa Fluor 647 antibody (EP4863(2))	-do-	ab246730
Goat polyclonal anti-melanoma gp100 (*C*-term)	Origene	TA302608
Rabbit monoclonal anti-tyrosinase antibody (369-377)	GenScript	RP10270
Anti-β-actin IgG antibody, mouse monoclonal (AC-15)	Sigma-Aldrich	A1978
Mouse IgG1 kappa isotype control	Invitrogen	14-4714-85
Goat anti-mouse IgG (H+L) Alexa Fluor 488	-do-	A32723
Goat anti-rabbit IgG (H+L) Alexa Fluor 488	-do-	A32731
Rabbit polyclonal anti-MART-1	-do-	PA5-120982
Cy5 affinipure donkey anti-mouse IgG (H+L)	Jackson Laboratories	715-175-150
Recombinant proteins and lectins		
FITC-avidin	Sigma	A2901
Recombinant Siglec E/Fc chimera, sourced from a mouse myeloma cell line, NS0-derived mouse Siglec-E protein	R&D systems	5806-SL
Biotinylated *Maackia amurensis*-II (MAL-II)	Vector Laboratories	B-1265
Fluorescein isothiocyanate conjugated *Sambucus nigra* agglutinin (FITC-SNA)	-do-	FL-1301
Fluorescein isothiocyanate conjugated *Vicia villosa* agglutinin (FITC-VVA)	-do-	FL-1231
Fluorescein isothiocyanate conjugated wheat germ agglutinin (FITC-WGA)	-do-	FL-1021
Neuraminidase from *Clostridium perfringens* expressed in *E. coli*	New England Biolabs	P0720L
Chemicals and other resources		
Doxorubicin HCl	Goldbio	D-490-500

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Dulbecco's modified Eagle's medium (DMEM)	Gibco	11965-092
Fetal bovine serum (FBS)	-do-	10082-147
Penicillin-Streptomycin (Pen-Strep)	-do-	15140-122
0.25 % Trypsin-EDTA	-do-	25200-056
Blasticidin S HCl	GoldBio	B-800-100
Protease inhibitor cocktail (PIC)	Sigma-Aldrich	P8340
radioimmunoprecipitation assay (RIPA) buffer	-do-	R0278
Propidium iodide (PI)	-do-	P4170
(3-(4,5-Dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT)	-do-	M2128
DAPI	-do-	D9542
Rhodamine phalloidin	Invitrogen	R415
BSA	-do-	A7030
Gelatin	-do-	48722
Hydrochloric acid	-do-	30721
N-acetyl-D-neuraminic acid	-do-	A0812
Ammonium persulfate	-do-	A3678
Dithiothreitol (DTT)	-do-	43819
Tris(2-carboxyethyl) phosphine hydrochloride (TCEP)	-do-	C4700
β-mercaptoethanol	-do-	M7522
Dimethyl sulfoxide (DMSO)	Fisher Scientific	67-68-5
Melanin	Merck	M0418
Sodium hydroxide	-do-	1.93102.0521
Sodium chloride	-do-	1.93206.0521
Sodium thiosulfate pentahydrate	-do-	61778405001730
Sodium carbonate anhydrous	Qualikems	S039112
Di-sodium hydrogen phosphate dihydrate	Merck	1.93622.0521
Potassium dihydrogen phosphate	-do-	7778-77-0
Potassium chloride	-do-	7447-40-7
Tris(hydroxymethyl)aminomethane	-do-	1.93315.0521
Glycerol anhydrous	-do-	1.07051.0521
Glycine	-do-	1.94907.0521
Tween 20,	-do-	8.22184.0521
Paraformaldehyde powder	-do-	158127
Chloroform	-do-	60244410001730
2-propanol	-do-	1.94524.0521
Methanol	-do-	60600725001730
Glacial acetic acid	-do-	1.93402.0521
37 % aq. Formaldehyde	-do-	1.94989.0521
Ethanol absolute	-do-	64-17-5
Crystal violet	-do-	C0775
RT^2 Profiler PCR Array Mouse Tumor Metastasis kit	Qiagen	PAMM-028ZA-6
Nuclease-free water	-do-	1039498

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Carboxyfluorescein succinimidyl ester (CFSE)	Invitrogen	C1157
TRIzol reagent	-do-	15596-026
SuperScript III First-Strand Synthesis System	-do-	18080051
eFluor 506 fixable viability dye	-do-	65-0866-14
ProLong Gold Anti-fade	Life Technologies	P36934
Skim milk powder	HiMedia	GRM1254
Bradford reagent	Bio-Rad	5000006
40 % aq. Acrylamide solution	-do-	1610140
2 % aq. Bis acrylamide solution	-do-	1610142
Precision Plus Protein Dual Color Standards	-do-	1610374
Bromophenol blue electrophoresis purity reagent	-do-	161-0404
Triton X-100 Detergent	-do-	1610407
Ethidium Bromide solution	-do-	161-0433
Sodium dodecyl sulfate (SDS)	SRL Chem	14374
Anhydrous sodium carbonate	-do-	1949157
N, N, N', N' Tetramethyl-ethylenediamine (TEMED)	MP Biochemicals	194019
Agarose Plus	Pronastar	8045
1kb DNA ladder	New England Biolabs	N3232S
6× Gel Loading Dye (blue)	-do-	B7021S
KAPA SYBR FAST	KAPA biosystems	KK4602
DBCO-Cy5	Click Chemistry Tools	A130-25
Maleimide-PEG_3-biotin	-do-	1029
Silver nitrate	BDH	I132529
D-Luciferin potassium salt	GoldBio	LUCK
Ketamine (80 mg/kg)	Zokent, Miracalus Pharma Private Limited	--
Xylazine (20 mg/kg)	Indian Immunologicals Limited	--
0.2 µm syringe PTFE filters	Millex-LG	SLLG013SL
PET membrane inserts 24 well 8.0 µm pore size	Corning	353097
Matrigel coated insert 24 well 8.0 µm pore size	-do-	354480
1 mL Syringe (26G ½)	BD	301321
Digital vernier calipers	SKADIOO	--

Supplemental Material

Supplementary material includes supporting results and discussion (in the main text), and figures S1−S5 with corresponding legends (in the supplementary section).

Data availability

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The data supporting the findings of this study are available from the corresponding author upon reasonable request.

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Author contributions

TJ, AR, and SGS conceptualized the study, designed the experiments, analyzed and interpreted the data, and wrote the manuscript. TJ, AR, and SGK performed the experiments *in vitro*. AR performed experiments *in vivo* in mice. SS provided critical inputs and the cell lines. NP trained and supervised the experiments in mice. SGS directed and supervised the research. All authors reviewed and approved the final manuscript.

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Competing interest

The authors declare no competing interests.

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Supplementary Material

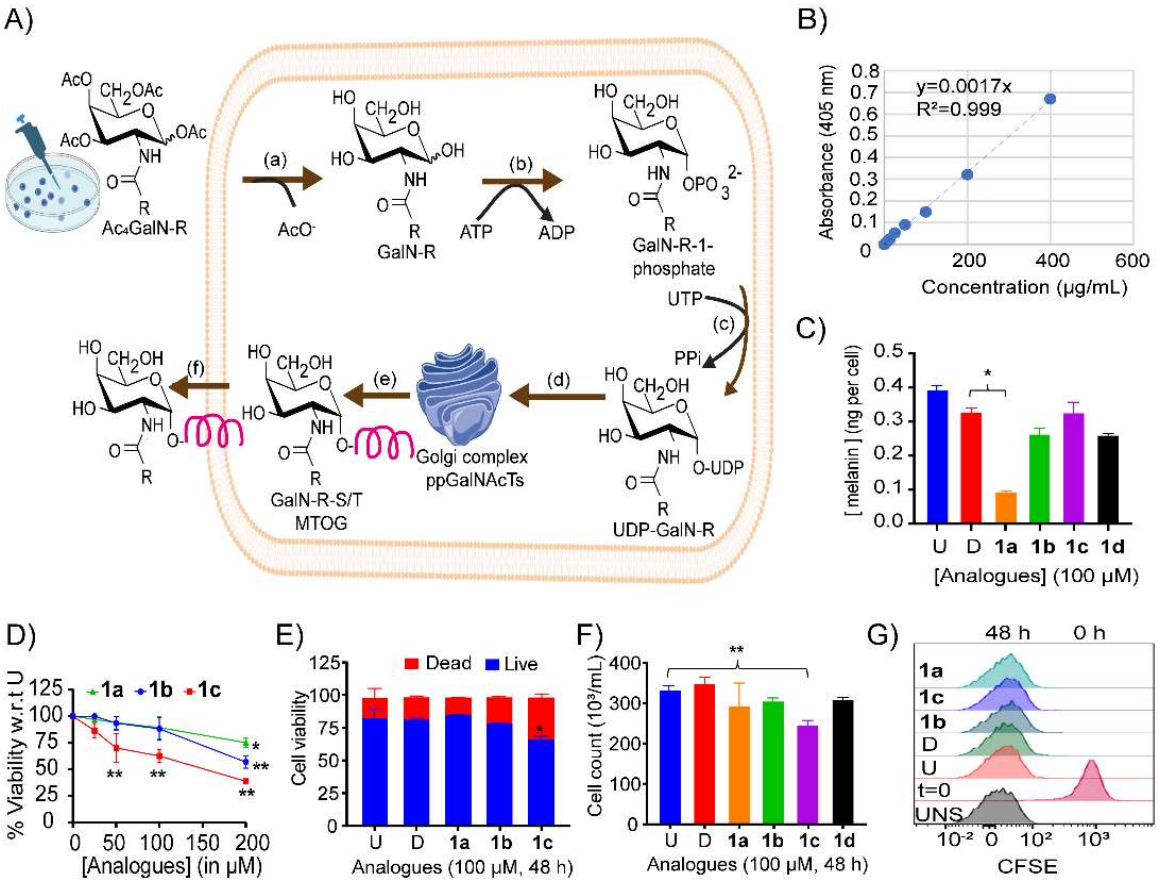


Figure S1. Metabolic glycan engineering (MGE) by GalNAc analogues selectively suppresses melanogenesis without affecting cell viability or proliferation. (A) Schematic illustrating MGE by peracetylated GalNAc analogues. (a) Peracetylated analogues passively diffuse into cells and are deacetylated by intracellular esterases. (b) The resulting free analogue undergoes phosphorylation at the C1-OH position by GalNAc-1-kinase. (c) Phosphorylated intermediates are activated to UDP-GalNAc analogues by UDP-GalNAc pyrophosphorylase. (d) These donors are transported into the Golgi lumen via the UDP-GalNAc transporter and are subsequently utilized by the mucin-type *O*-glycosylation enzymatic machinery. (e) This results in the incorporation of modified GalNAc analogues into *O*-glycans. (f) The modified *O*-glycans are expressed on cell surface glycoproteins, enabling detection and functional analyses. (B-C) Effect of GalNAc analogues on melanin biosynthesis. (B) Melanin standard curve generated using synthetic melanin solutions ranging from 0 to 400 µg/mL. (C) Quantification of intracellular melanin content in B16F10 cells upon analogue treatment. Data is presented as melanin concentration (ng per cell). Error bars represent SD of three BR. Statistical significance was determined using one-way ANOVA. P-values are *<0.05 compared with vehicle D. (D-G) Effect of GalNAc analogues on B16F10 cell viability and proliferation. (D) Cell viability assessed by MTT assay upon analogue treatment (0-200 µM), expressed as percentage relative to untreated cells. Seeding density: 10,000 cells/well (E) Percentage of dead and live cells determined by trypan blue exclusion following analogue treatment. Approximately 100 cells per sample were counted using a hemocytometer. (F) Cell density (× 10^3/mL) measured after 48 h upon analogue treatment at 100 µM. Cells were plated at 5.0 × 10^4/mL in 10 mL of DMEM. (G) CFSE dye dilution assay assessing cell proliferation. Representative histograms show CFSE fluorescence at 0 h and dilution at 48 h upon analogue treatment (100 µM, 48 h) in B16F10 cells. Error bars represent SD of three BR in (D-F) and two BR (each sample counted twice for 10,000 PI-negative events) in (G). Statistical analysis was performed using two-way ANOVA for MTT, Trypan blue, and CFSE assays, and one-way ANOVA for cell count. P-values are *<0.05, **<0.01, compared to vehicle. U, untreated; D, DMSO (vehicle); **1a**, Ac_5GalNTGc; **1b**, Ac_4GalNAc; **1c**, Ac_5GalNGc; and **1d**, Ac_3BnGalNAc.

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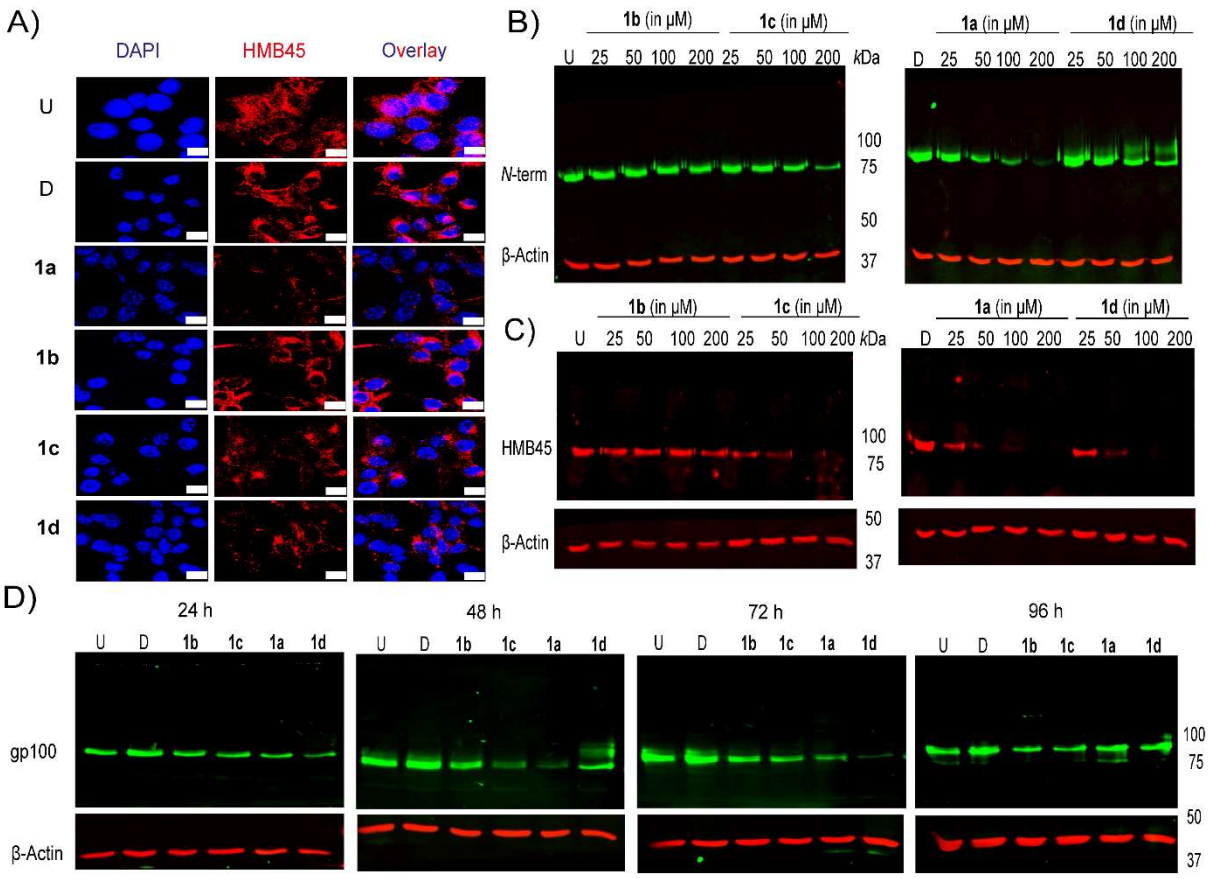


Figure S2. Dose-dependent and time-course modulation of Pmel17/gp100 glycosylation by GalNAc analogues. (A) Confocal imaging of B16F10 cells treated with GalNAc analogues (100 μM, 48 h), fixed, permeabilized, and stained with anti-gp100 antibody (HMB45, red), along with DAPI (blue). Imaging was performed using 63× oil immersion on a Zeiss LSM 980 confocal microscope. Images shown are representative of three BR with two technical replicates per BR. Each replicate was conducted on one cover slip with five fields imaged per coverslip. Scale bar: 5 μm. (B-C) B16F10 cells were treated with GalNAc analogues (0-200 μM, 48 h). Cells were lysed, and cell lysates were subjected to western blotting using: (B) anti-gp100 (*N*-term) antibody; (C) anti-gp100 (HMB45) antibody, which specifically recognizes a glycan-dependent epitope on Pmel17/gp100. The blots showed dose-dependent loss of HMB45 reactivity upon **1a** treatment. (D) Time-course analysis of Pmel17/gp100 modulation. B16F10 cells were treated with GalNAc analogues (100 μM) and lysed at respective time points (24, 48, 72, and 96 h). Cell lysates were probed with anti-gp100 (*N*-term) antibody. Western blots showed maximal inhibition of Pmel17/gp100 levels at 48 h upon **1a** treatment, with recovery of Pmel17/gp100 levels by 96 h, consistent with the reversible nature of MGE. β-actin blots are shown as loading controls. Blots shown are representative of two BR with two technical replicates per BR. U, untreated; D, DMSO (vehicle); **1a**, Ac_5GalNTGc; **1b**, Ac_4GalNAc; **1c**, Ac_5GalNGc; and **1d**, Ac_3BnGalNAc.

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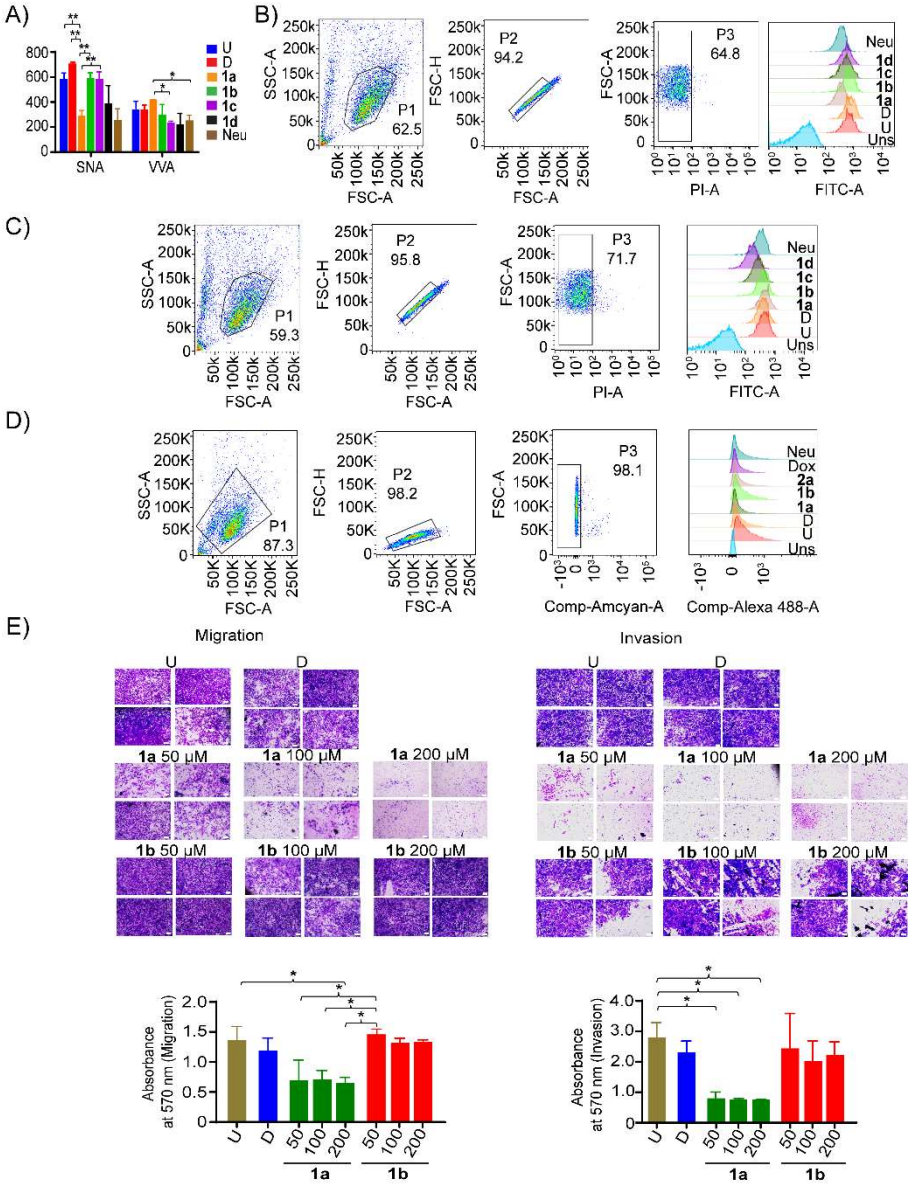
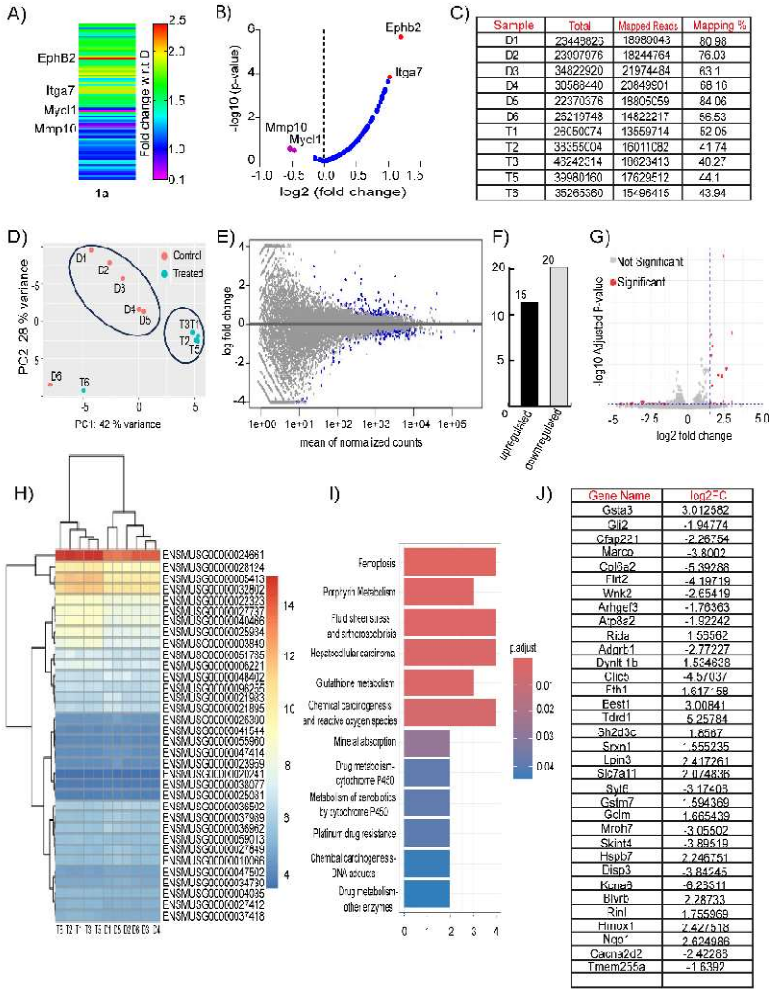


Figure S3. 1a-mediated decrease in cell-surface sialylation, Siglec ligand engagement, and metastatic traits in melanoma cells. (A-C) Quantification of cell-surface sialic acid levels in B16F10 cells treated with GalNAc analogues. (A) B16F10 cells treated with GalNAc analogues (100 µM, 48 h) were stained with fluorescein-conjugated SNA (0.5 µg/mL) and VVA lectin (10 µg/mL) for 15 min at RT. Cells treated with α2→3, 6, 8-neuraminidase from *Clostridium perfringens* (100 units/mL) for 30 min were used as negative controls. Stained cells were analyzed by flow cytometry, and GMFI of FITC-positive populations was quantified and plotted as bar graphs. Error bars indicate the SD of two BR with two technical replicates each. Statistical significance was determined using two-way ANOVA with Tukey's multiple comparisons test. P-value * ≤0.05, ** ≤0.01, and all other comparisons were non-significant. (B-C) Flow cytometry gating strategy for lectin binding analysis. Cells were sequentially gated on SSC-A vs FSC-A (to exclude debris), FSC-H vs FSC-A (to select single cells), and FSC-A vs PI-A (to select live cells). Histograms represent FITC-positive populations for (B) SNA binding and (C) VVA binding. (D) Gating strategy for Siglec-E-Fc binding. Cells were sequentially gated on SSC-A vs FSC-A, FSC-H vs FSC-A, and FSC-A vs Comp-AmCyan-A to select single live cells. Histograms depict AF488-positive populations corresponding to Siglec-E-Fc binding. (E) **1a**-treatment suppresses melanoma cell migration and invasion in a dose-dependent manner. B16F10-Luc2 (1.0×10^5) cells were treated with vehicle (DMSO), **1a**, or **1b** at 50, 100, or 200 µM concentrations and seeded into transwell inserts. For invasion assays, inserts were coated with Matrigel to mimic the extracellular matrix. After monolayer formation (~12 h), a serum gradient was introduced, and cell migration or invasion through an 8.0 µm PET membrane was assessed after 48 h. Representative images of cells that migrated or invaded to the bottom side of the membrane, fixed and stained with 0.25 % crystal violet, are shown. [Scale bar: 100 µm, images representative of four fields per insert, 10× objective]. Cells on the lower surface were solubilized in DMSO, and absorbance was measured at 570 nm. Error bars represent the SD of two BR. Statistical significance was determined using one-way ANOVA with Tukey's multiple comparisons test. P-value * ≤ 0.1 for invasion and P-value * ≤ 0.05 for migration. U, untreated; D, DMSO (vehicle); **1a**, Ac_5GalNTGc; **1b**, Ac_4GalNAc; **1c**, Ac_5GalNGc; **1d**, Ac_3BnGalNAc; **Neu**, neuraminidase; **2a**, Ac_5GlcNTGc; **Dox**, Doxorubicin; IC, isotype control; and Uns, Unstained.

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Sample	Total	Mapped Reads	Mapping %
D1	23443823	18080043	80.98
D2	23007076	18244764	76.03
D3	34822920	21974484	63.1
D4	30568440	20649801	68.16
D5	22370376	18805059	84.06
D6	25219748	14822217	58.53
T1	28050074	13559714	52.06
T2	33555004	16011082	47.74
T3	43242314	19823413	40.27
T5	39980160	17629512	44.1
T6	35265360	15496415	43.94

Gene Name	log2FC
Gsta3	3.012582
Gil2	-1.94774
Cfap221	-2.26754
Marco	-3.8002
Col3a2	-5.39288
Flrt2	-4.19719
Wnk2	-2.65419
Arhgef3	-1.76363
Arhgef2	-1.92242
Rida	1.53562
Adgrl1	-2.77227
Dynlt1b	1.534838
Clic5	-4.57037
Fth1	1.617758
Eest1	3.00841
Tdrd1	6.25784
Sh2d3c	1.6567
Scyl1	1.555235
Lpin3	2.417261
Slc7a11	2.074836
Sytl1	-3.17409
Gstm7	1.594369
Golm	1.665439
Mroh7	-3.05502
Skint4	-3.89519
Hspb7	2.246751
Disp3	-3.84545
Kcna3	-6.23511
Blvrb	2.28733
Rin1	1.755969
Hmox1	2.427518
Ngp	2.624986
Cacna2d2	-2.42283
Tmem255a	-1.6392

Figure S4. Targeted metastasis gene profiling reveals selective transcriptional effects of 1a, while global transcriptomic analysis indicates minimal off-target gene regulation. (A-B) Changes in mRNA levels of metastasis-associated genes in B16F10 cells following **1a** treatment assessed using a mouse metastasis RT^2 Profiler PCR array. (A) Heatmap showing fold changes in metastatic genes expression upon **1a** treatment (100 μM, 48 h) relative to the vehicle (DMSO). Four significantly altered genes are highlighted: EphB2 and ITGA7 were upregulated by 2-fold, while MMP-10 and Mycl-1 were downregulated by 0.5-fold upon **1a** treatment. (B) Volcano plot shows relative expression changes, depicted as log2(fold change), plotted against statistical significance -log10(P-value). Red indicates genes significantly upregulated by ≥2-fold. Pink indicates genes significantly downregulated by ≥2-fold upon **1a** treatment. Data represent n = 2 BR run on two independent PCR array plates. (C-J) Transcriptomic profiling of B16F10 cells treated with **1a** reveals minimal global gene expression changes. (C) Summary of sequencing read statistics, including total reads, mapped reads, and mapping efficiency for all qualified samples. Sample T4 was excluded due to low RNA quality (Status: Fail). (D) Principal Component Analysis (PCA) plot showing distinct clustering between the control (DMSO, red) and treated (**1a**, blue) groups. (E) MA plot visualizing log2 fold changes versus mean normalized expression levels. (F) Bar graph summarizing differentially expressed genes (DEGs), with 15 upregulated and 20 downregulated genes. (G) Volcano plot displaying DEGs; red dots represent genes meeting the significance threshold (adjusted P-value < 0.05 and |log2FC| > 1.5), while grey dots represent non-significant genes. (H) Heatmap with hierarchical clustering of significant DEGs across all samples. (I) Pathway enrichment analysis (KEGG/GO) of DEGs, highlighting enrichment of pathways related to ferroptosis, porphyrin metabolism, and glutathione metabolism. Bars are colored based on adjusted P-value. (J) Table of DEGs listing gene name and log2 fold change (log2FC). D, DMSO (vehicle) and **1a**, Ac_5GalNTGc.

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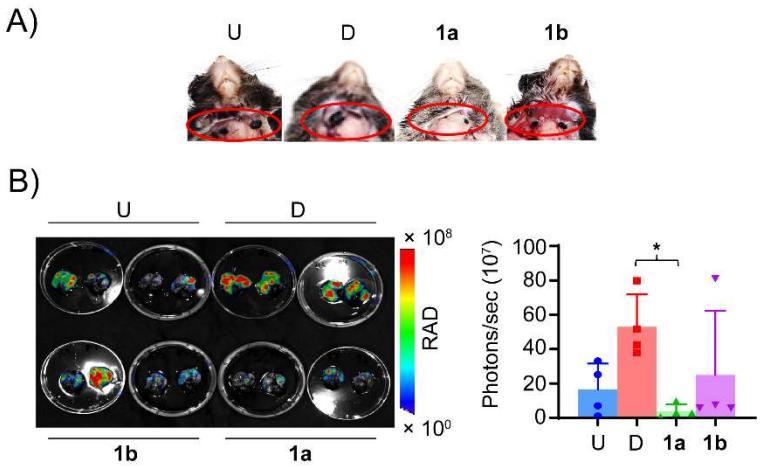


Figure S5. Intravenous injection of 1a-pretreated melanoma cells slows down metastases. (A) Metastatic nodules were also observed in the lymph nodes surrounding the neck region following dissection. (B) *Ex vivo* bioluminescence imaging of lungs soaked in D-luciferin substrate was performed using a Lago X imager. BLI (photons/sec) was quantified using Aura software. Error bars represent the SD of n = 4 mice per group. Statistical significance was determined using the Kruskal-Wallis test followed by Dunn's multiple comparisons test. P-value * ≤ 0.05. U, untreated; D, DMSO (vehicle); **1a**, Ac_5GalNTGc; and **1b**, Ac_4GalNAc.